

A Digital Twin-Driven Framework for Low-Power DRAM Optimization Using PySpice and Machine Learning

A configuration-driven, end-to-end framework that couples automated NGSpice characterisation of a 1T1C DRAM bit cell with machine-learning surrogates, a deployable digital twin, robust multi-objective optimisation and explainable-AI analysis. Every reported number is produced by the accompanying code from executed simulations.

Generated 2026-07-23T13:46:25+00:00

Contents

- [1. Introduction](#)
- [2. Related work \(2021-2026\)](#)
- [3. Framework](#)
- [4. Experimental setup](#)
- [5. Dataset and physical validation](#)
- [6. Surrogate models and the digital twin](#)
- [7. Multi-objective optimisation](#)
- [8. Explainability, robustness and assist techniques](#)
- [9. Limitations and threats to validity](#)
- [10. Reproduction](#)

1. Introduction

DRAM bit-cell design is a genuinely multi-objective problem. Retention time, read latency, read margin and power cannot be improved independently: raising the storage capacitance improves retention and sense margin but costs area and access energy; boosting the wordline enables a full '1' write but increases both power and gate-oxide stress; lowering the bitline precharge level saves precharge energy but erodes the margin for one of the two data polarities. Exploring that space directly with a circuit simulator is expensive, and the expense grows multiplicatively once process corners, temperature and statistical variation enter.

This work builds a complete framework that makes such exploration tractable. A parameterised 1T1C cell, its bitline pair, boosted precharge network, CMOS write driver and cross-coupled sense amplifier are generated from configuration and characterised in NGSpice. A space-filling design of 50000 points is simulated, machine-learning surrogates are fitted to the resulting responses, and the best surrogate per response is deployed as a digital twin. The twin then drives a robust multi-objective search whose solutions are returned to the circuit simulator for verification.

What is contributed. (i) A configuration-driven pipeline in which no DRAM architectural constant is hard-coded, so retargeting to another cell or technology node means editing YAML, not Python. (ii) A retention model that is *measured* rather than assumed: the storage-node leakage characteristic is extracted by a DC

sweep of a leakage replica inside the same simulation, and the charge-loss ODE is integrated over it -- with the result cross-checked against direct long-window transient simulation. (iii) A broad, statistically tested comparison of 14 surrogate families and six optimisers under identical budgets. (iv) A published dataset of every simulated design point. (v) Explainability and robustness analyses that turn the surrogate from a black box into a source of design guidance.

What is deliberately not claimed. The device models are the public PTM BSIM4 cards for a general-purpose low-power logic process, not a DRAM-specific process. Absolute retention times therefore reflect those cards and are not predictions about any commercial part; the framework, the relative trends and the methodology are the contribution. Section 9 states the limitations in full.

2. Related work (2021-2026)

Machine learning for analog and mixed-signal design has converged on two families: *simulation-in-the-loop* methods, which call the simulator inside the optimiser, and *surrogate-based* methods, which replace most of those calls with a learned model. Recent surveys of the area place both families side by side, and recent optimisation studies report reductions in SPICE calls of roughly 56-83 %, with multi-fidelity schemes reporting an order-of-magnitude reduction in total simulation time while still matching ground-truth Pareto fronts. This framework sits in the surrogate family and adopts the same discipline of returning to the simulator for verification.

On the tooling side, programmatic SPICE deck generation with PySpice has recently been used both to build datasets for learning and to drive complete simulation studies, which supports the toolchain adopted here. On the DRAM side, published work concentrates on the architecture and system levels -- refresh scheduling, retention profiling, data mapping and measurement infrastructure -- and on demonstrating that DRAM modelling accuracy must be validated rather than assumed. The present study works one level below those, at the bit cell and its column periphery, and adopts the same validation discipline: the retention model is checked against direct transient simulation before it is used.

The gap this work addresses is the absence of an *end-to-end, reproducible and openly-data-backed* pipeline that takes a configurable DRAM cell from netlist generation through surrogate modelling, digital-twin deployment, robust multi-objective optimisation, explainability and statistical validation.

Table 1. Recent works positioning this study.

Work	Year	Focus
Multi-Fidelity Surrogate Models for Accelerated Multi-Objective Analog Circuit	2025	surrogate modelling; multi-objective circuit optimisation
Analog Design and Machine Learning: A Review	2025	review of ML in analog design
Analog circuit sizing based on Evolutionary Algorithms and deep learning	2024	evolutionary sizing with learned models
Machine Learning Driven Global Optimisation Framework for Analog Circuit Design	2024	ML-guided global optimisation
Performance Evaluation of Evolutionary Algorithms for Analog Integrated Circuits	2023	benchmarking evolutionary optimisers on circuit problems
The Dawn of AI-Native EDA: Opportunities and Challenges of Large	2024	AI-native EDA landscape

Circuit Model		
When Device Modeling Meets Machine Learning: Opportunities and Challenges	2024	ML for compact/device modelling
Digital Twin for Secure Semiconductor Lifecycle Management: Prospects and Appl	2022	digital twins in the semiconductor lifecycle
SPICEPilot: Navigating SPICE Code Generation and Simulation with AI Guidance	2024	automated SPICE deck generation, PySpice-based datasets
Python Framework for Modular and Parametric SPICE Netlists Generation	2023	parametric netlist generation
PySpice-Simulated In Situ Learning with Memristor Emulation for Single-Layer S	2024	PySpice/NGSpice in a research pipeline
EasyDRAM: An FPGA-based Infrastructure for Fast and Accurate End-to-End Evalua	2025	DRAM characterisation infrastructure
Re-Evaluating the Real-System Modeling Accuracy of Ramulator 2.0	2026	DRAM simulator fidelity
PENDRAM: Enabling High-Performance and Energy-Efficient Processing of Deep Neu	2024	DRAM energy optimisation at the architecture level

Full bibliographic records and retrieval URLs are in docs/literature/references.yaml.

3. Framework

The framework follows the seven stages of the proposed method.

Stage 1 -- cell design and parameterisation. `dramdt.models.cell` emits the BSIM4 deck for a 1T1C cell, a lumped bitline pair, a boosted precharge/equalise network, a CMOS transmission-gate write driver and a cross-coupled sense amplifier. Two design details are worth stating because they are easy to get wrong and both were found by simulation during development: the precharge pass devices must be driven from the boosted rail, because a VDD-level gate leaves $V_{gs} = V_{DD}/2 < V_{th}$ and cannot pull the bitline to $V_{DD}/2$; and the access device needs $V_{PP} > V_{DD} + V_{th,eff}$ to store a full '1', which is why the wordline-boost range extends to 1.2 V.

Stage 2 -- automated simulation. Each deck runs three analyses in one NGSpice invocation: an operating point for standby current, a transient covering write / precharge / read / sense, and a DC sweep of a *leakage replica* -- an identical access device whose storage node is driven by a probe source -- yielding the storage-node leakage characteristic. Tolerances are set per analysis, because a single ABSTOL cannot resolve femto-ampere leakage and micro-ampere switching in the same deck.

Stage 3 -- dataset. Latin hypercube sampling over 15 variables, decoded through the configuration's derived expressions. Design points that fail to write or to sense are *retained and labelled*, not discarded: a surrogate that has never seen an infeasible design cannot steer an optimiser away from one.

Stage 4 -- surrogates. A zoo spanning linear, kernel, neighbour, ensemble, boosting and neural families is cross-validated on train+val; the test split is touched once per model.

Stage 5 -- optimisation. Six optimisers under an identical evaluation budget, each repeated 7 times. Objectives and constraints are reduced pessimistically over the configured PVT conditions, making the search a robust-design problem.

Stage 6 -- digital twin. The best surrogate per response is packaged with the full input-preparation pipeline behind one predict call.

Stage 7 -- analysis. Assist-technique comparison, SHAP/ALE explainability, Monte-Carlo yield, corner sweeps and Sobol' sensitivity.

The pseudocode for the six core algorithms is in `reports/algorithms/`.

4. Experimental setup

Item	Value
Simulator	NGSpice
Device models	PTM BSIM4 (level 54), ? nm
Model-card fingerprint	?
Process corners	
Design variables	15
Samples	50000
Sampler	lhs
Master seed	20260723
Configuration hash	e22061fdb47e16e7
Platform	?

Corner cards for SS/FF/SF/FS are synthesised from the typical PTM card by the first-order $v_{th0}/u_{0}/t_{ox}$ shifts declared in the configuration; they are explicitly *not* foundry sign-off corners. A solver convergence study (Table: `solverconvergencestudy`) shows the production setting differs from a 20 ps GEAR reference by at most 0.70 % at the 95th percentile on every metric, with pass/fail labels unchanged.

All randomness derives from the single master seed through a pure (seed, label, index) hash, so results are independent of worker count and completion order.

5. Dataset and physical validation

The campaign completed **50000 simulations** (71428027540.329 per second on 18 workers, 0.0 min total), of which 49999 produced measurements (100.00 %). Preprocessing retained 49987 of 50000 rows (99.97 %); the removals were {'simulationfailed': 1, 'notconverged': 3, 'incomplete_measurements': 9}. 57.9 % of the retained points are feasible (both write and read succeed); the remainder are kept and labelled as infeasible.

The Latin hypercube achieved a centred-L2 discrepancy of 0.00680435 and a minimum pairwise distance of 0.4115 (both on a 4000-point subsample).

Two independent physics checks were run before trusting the dataset.

First, the measured bitline differential was compared with the analytical charge-sharing law $dV = (V_{SN} - V_{BLpre}) * C_s / (C_s + C_{b1})$: median relative error **3.802 %**, 95th percentile 17.992 % over 120 design points.

Second, the quasi-static retention model was checked against direct long-window transient simulation on 200 designs: Pearson $r = \mathbf{0.99974}$ in log10 space, Spearman $\rho = 0.9995$, median absolute error **0.0107 decades**,

and 100.0 % of designs within a factor of two. The retention numbers in this study are therefore integrations of a measured leakage characteristic whose accuracy has been quantified, not an analytical shortcut.

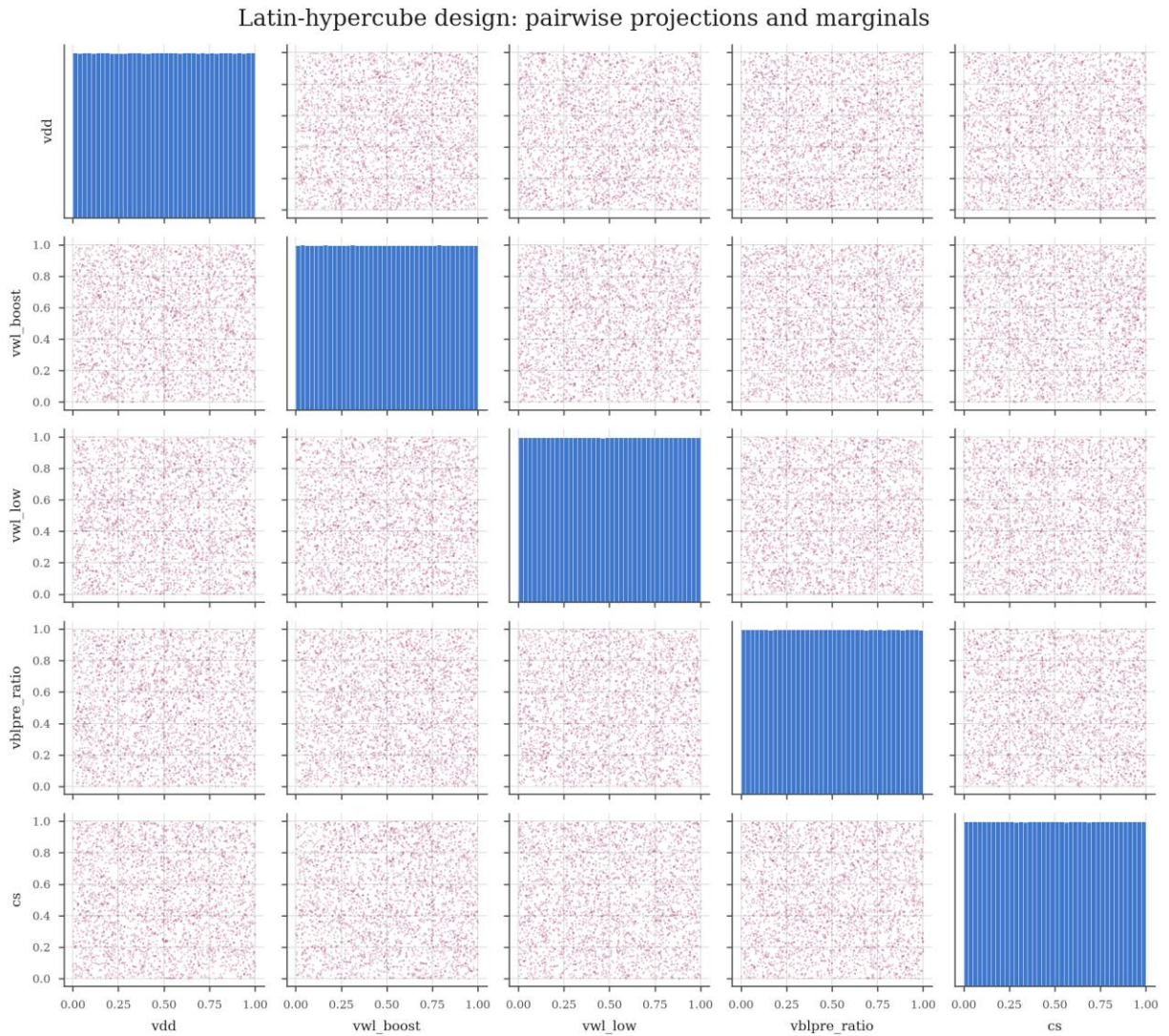


Figure 1. Latin-hypercube design: pairwise projections and marginals.

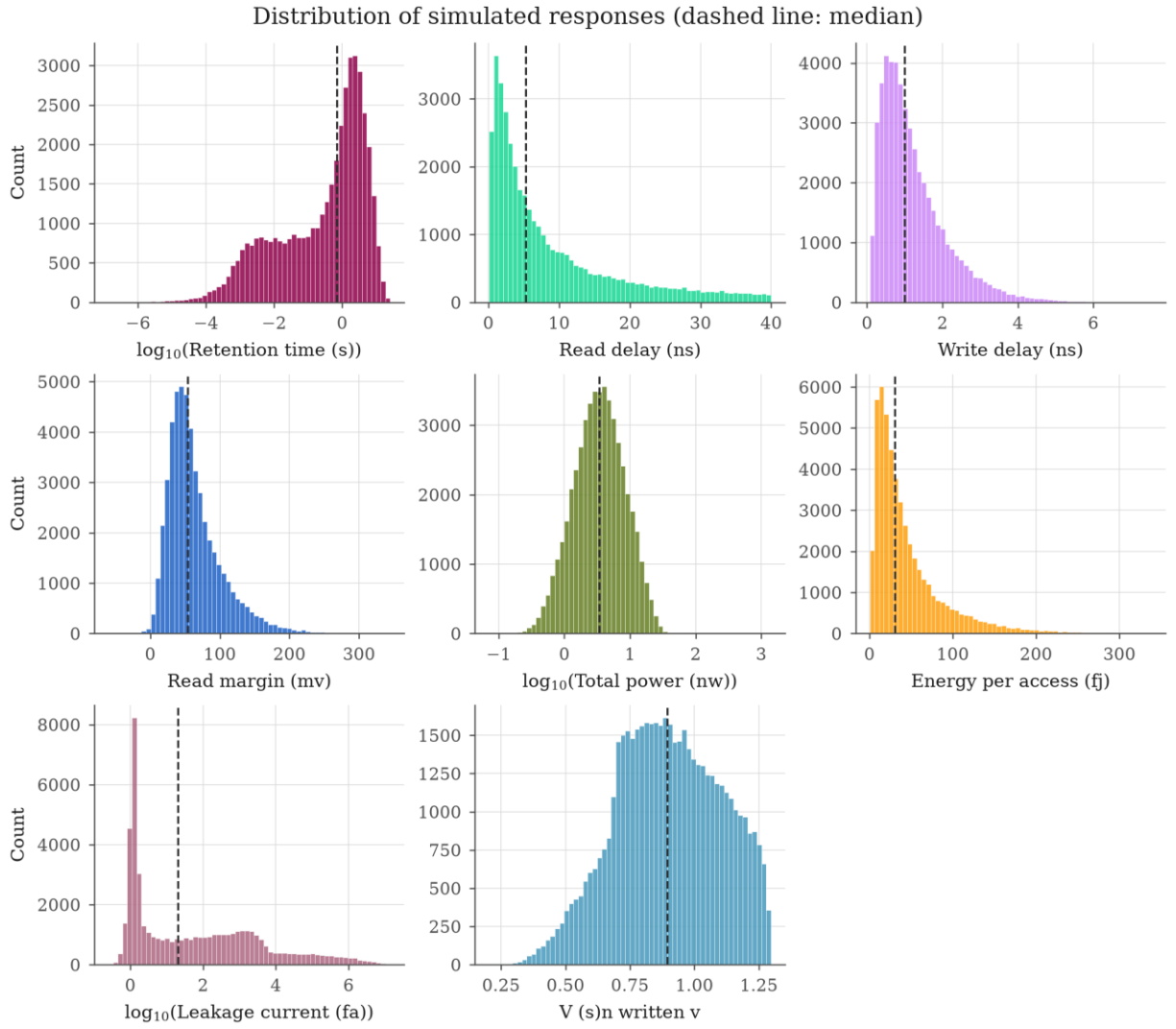


Figure 2. Distribution of the simulated responses.

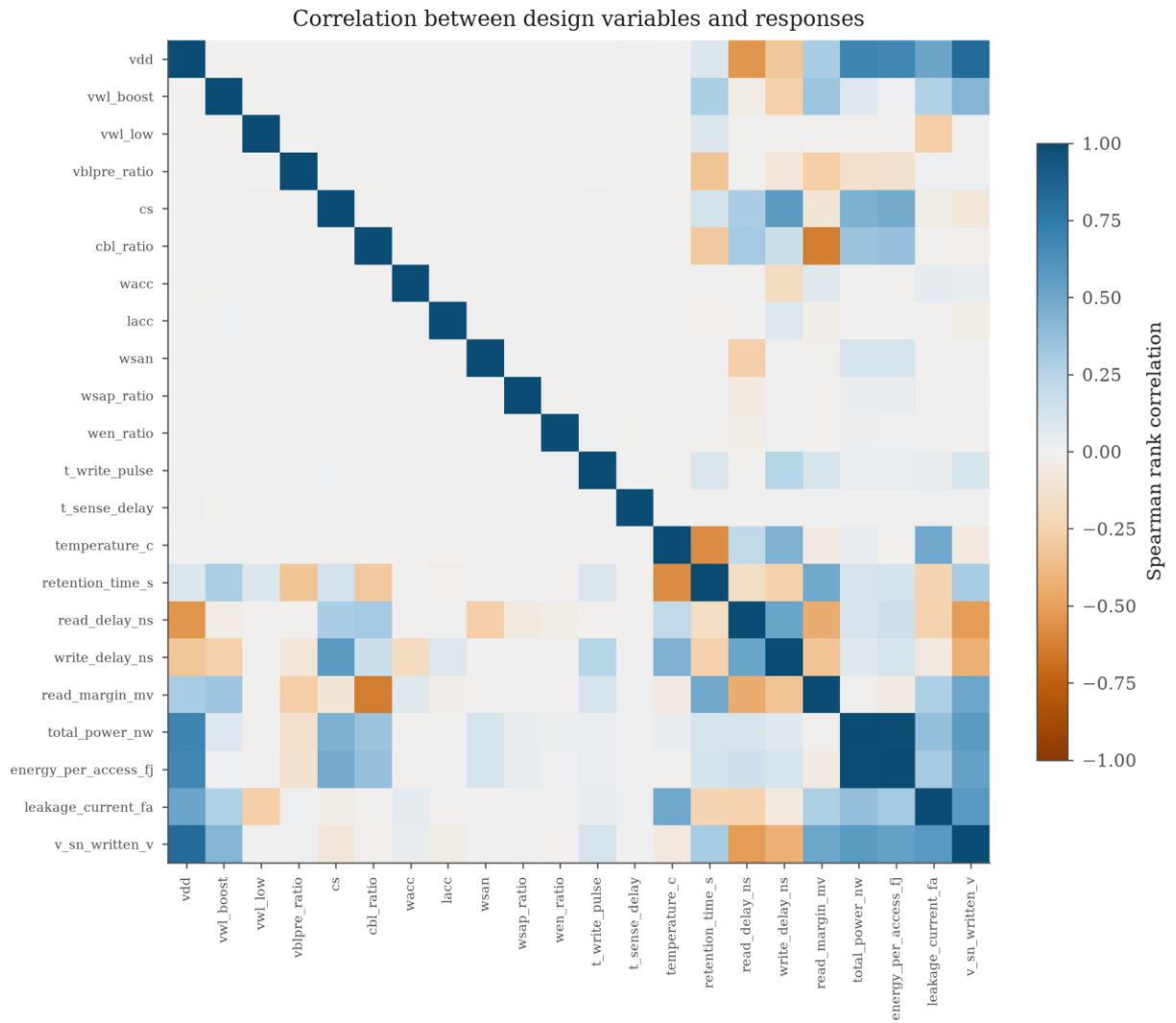


Figure 3. Rank correlation between design variables and responses.

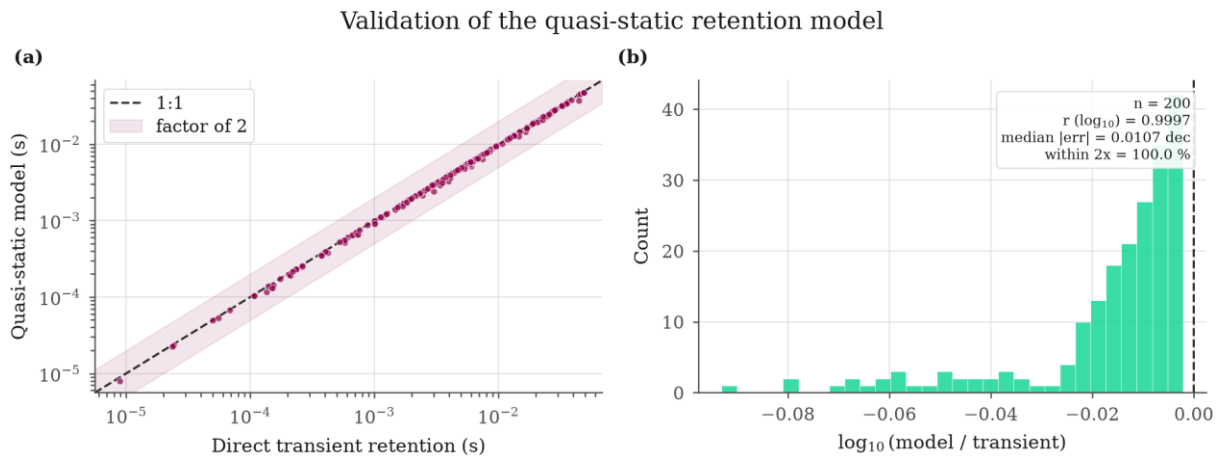


Figure 4. Validation of the quasi-static retention model against direct transient simulation.

Table 2. Measured bitline differential against the charge-sharing law.

sample	corner	temperature_c	cs_ff	cbl_ff	vsnwritten_v	measureddvmv	analyticaldvmv	relativeerrorpct
--------	--------	---------------	-------	--------	--------------	--------------	----------------	------------------

0	TT	79.61	23.49	198.7	0.7813	45.07	44.86	0.4685
1	TT	-3.646	11.11	45.92	0.5546	48.81	47.59	2.568
2	TT	-7.444	9.672	109.0	1.098	48.88	47.21	3.536
3	FF	-38.56	29.77	171.6	0.8018	53.41	52.43	1.880
4	TT	107.9	7.442	76.53	0.9565	51.70	43.63	18.52
5	SF	-32.83	6.146	47.00	0.8195	57.01	51.13	11.49
6	SF	-17.01	10.88	109.1	0.8089	33.23	30.24	9.903
7	TT	55.37	23.79	52.85	0.8160	133.5	131.7	1.342
8	TT	73.93	7.127	33.60	1.199	132.5	125.2	5.789
9	SF	105.0	22.11	263.0	1.023	40.31	39.78	1.337
10	FS	36.83	27.30	143.9	1.198	101.0	101.8	0.8056
11	FF	29.88	29.98	189.1	0.9430	51.65	49.84	3.643
12	FS	60.77	6.861	16.15	0.6082	55.07	49.24	11.84
13	TT	-12.90	9.340	90.63	0.8712	31.93	30.82	3.627
14	SF	62.23	22.65	184.2	0.8458	58.98	57.44	2.687
15	FS	6.565	7.282	75.56	0.6259	23.95	23.33	2.682
16	SS	35.24	24.73	264.2	0.7510	14.23	14.02	1.493
17	SS	26.37	7.676	15.74	0.7849	112.1	115.9	3.254
18	FF	11.36	39.13	290.9	0.7774	44.35	43.67	1.576
19	SF	85.20	20.63	202.2	0.6425	24.96	13.87	79.99
20	FF	102.0	37.45	352.0	0.7889	12.61	12.16	3.704
21	SF	-34.42	8.131	77.02	1.010	55.09	50.40	9.315
22	TT	4.035	5.239	32.38	1.054	66.10	62.05	6.527
23	SF	23.43	10.68	59.74	0.6586	36.17	32.82	10.20
24	FF	-22.83	8.069	53.13	1.239	98.29	97.00	1.325
25	SF	8.580	26.03	90.16	0.6933	54.90	52.71	4.164
26	FS	-16.47	19.22	132.8	1.165	87.14	86.24	1.042
27	FS	0.8620	20.24	88.55	0.7496	57.41	55.76	2.973
28	TT	103.1	8.589	59.99	0.8577	42.46	37.28	13.89
29	FS	48.46	11.56	24.74	1.009	133.3	129.9	2.625
30	FS	-34.86	10.06	50.22	1.269	122.1	117.2	4.215
31	FF	-24.10	13.05	152.1	1.028	40.87	38.42	6.371
32	SS	15.50	5.412	52.07	0.6150	31.65	27.84	13.69
33	FF	67.51	8.909	73.46	0.7312	32.71	30.88	5.947
34	SS	-18.96	16.88	151.1	0.8777	33.45	33.55	0.3045
35	FF	113.0	28.36	235.7	0.8158	50.98	49.23	3.553
36	TT	-9.444	9.053	95.62	0.5893	18.31	17.07	7.227
37	SF	-6.491	12.07	51.77	1.059	83.05	78.52	5.777
38	FS	84.65	5.324	24.22	1.036	67.91	61.52	10.40
39	TT	107.0	16.36	63.41	1.071	111.2	106.8	4.091

Relative error between the simulated VBL - VBLB at the sense-amplifier firing instant and $(V_{SN} - V_{BLpre}) * C_s / (C_s + C_{bl})$.

Table 3. Agreement statistics for the retention model.

statistic	value
n_pairs	18
pearsonlog10	1.0000
spearman_rho	1
medianabslog10_error	0.009786

meanlog10error	-0.01160
geometricmeanratio	0.9736
p90abslog10_error	0.02026
withinfactor2_pct	100
withinfactor10_pct	100

Table 4. Solver convergence study.

s	m	sp	read	rea	rea	rea	writ	wri	vsn	vsn	rete	rete	ener	ene	total	tota	rea	writ
e	s	ee	mar	dm	ddel	dde	edel	ted	writ	wri	ntio	ntio	gype	rgyp	pow	lpo	dsu	esuc
t	p	du	gin	argi	ayn	lay	ayn	ela	ten	tte	ntim	nti	racc	erac	ern	wer	cces	cess
i	sa	pv	mv	nm	sme	nsp	sme	yns	vm	nv	esm	mes	essff	cess	wm	nw	sagr	agre
n	m	sre	med	vp9	dian	95e	dian	p95	edi	p9	edia	p95	medi	fjp9	edia	p95	eem	eme
g	pl	fer	iane	5err	err_	rr_	err_	err_	ane	5er	nerr	err_	aner	5err	nerr	err_	ent	nt_
e	ce	ce	ct	t	pct	pct	pct	pct	rrpct	rpc	_pct	pct	rpct	pct	_pct	pct	_pct	pct
50	63	1.693	0.009100	0.1449	0.01530	0.2420	0.1520	0.7592	0.003800	0.04200	0.001800	0.4649	0.05650	0.3348	0.05490	0.3300	100	100
50	60.90	1.752	0.003100	0.02840	0.01110	0.1547	0.03910	0.6964	7.000e-4	0.005800	3.000e-4	0.05170	0.02130	0.1653	0.02070	0.1591	100	100
100	47.40	2.249	0.01990	0.5967	0.1581	0.8344	0.3680	1.367	0.01190	0.1842	0.003400	1.738	0.1905	0.5836	0.1788	0.5444	100	100

Percentages are relative errors against the 20 ps GEAR setting over the same design points; agreement columns give the share of points whose pass/fail label is unchanged.

6. Surrogate models and the digital twin

The surrogate comparison is summarised below.

Response	Selected model	CV R2	Test R2	Test RMSE	Test MAE
write_efficiency	mlp_deep	0.99308	0.99859	0.004596	0.003023
energyperaccess_fj	mlp_deep	0.98961	0.99614	2.674	1.237
log10totalpower_nw	mlp_deep	0.97362	0.98266	0.05173	0.02065
readdelayns	mlp_deep	0.97108	0.98778	1.031	0.4237
readmarginmv	mlp	0.96139	0.98966	3.978	1.659
log10retentiontime_s	mlp_deep	0.91140	0.95098	0.9212	0.2123

The model in each row is the one selected by cross-validation on the train+val partition and deployed in the digital twin; the test columns are the single, untouched-until-then evaluation of that choice.

By family, the median held-out R2 ranks as: **neural** 0.9887, **boosting** 0.9635, **ensemble** 0.9041, **kernel** 0.8642, **neighbour** 0.7236, **linear** 0.7097.

Ranked by mean Friedman rank across every response (lower is better): *mlpdeep* (1.40), *mlp* (2.23), *catboost* (2.73), *lightgbm* (4.20), *histgbr* (4.83).

Digital twin cost. At a batch size of 10000 the twin evaluates a design in 0.03248 ms (30,786 designs/s), a **37,009x** speed-up over the measured mean SPICE evaluation time. This is what makes the optimisation, the Monte-Carlo yield study and the Sobol' analysis affordable.

Surrogate model comparison (bars: held-out test score, whiskers: 95 % bootstrap CI)

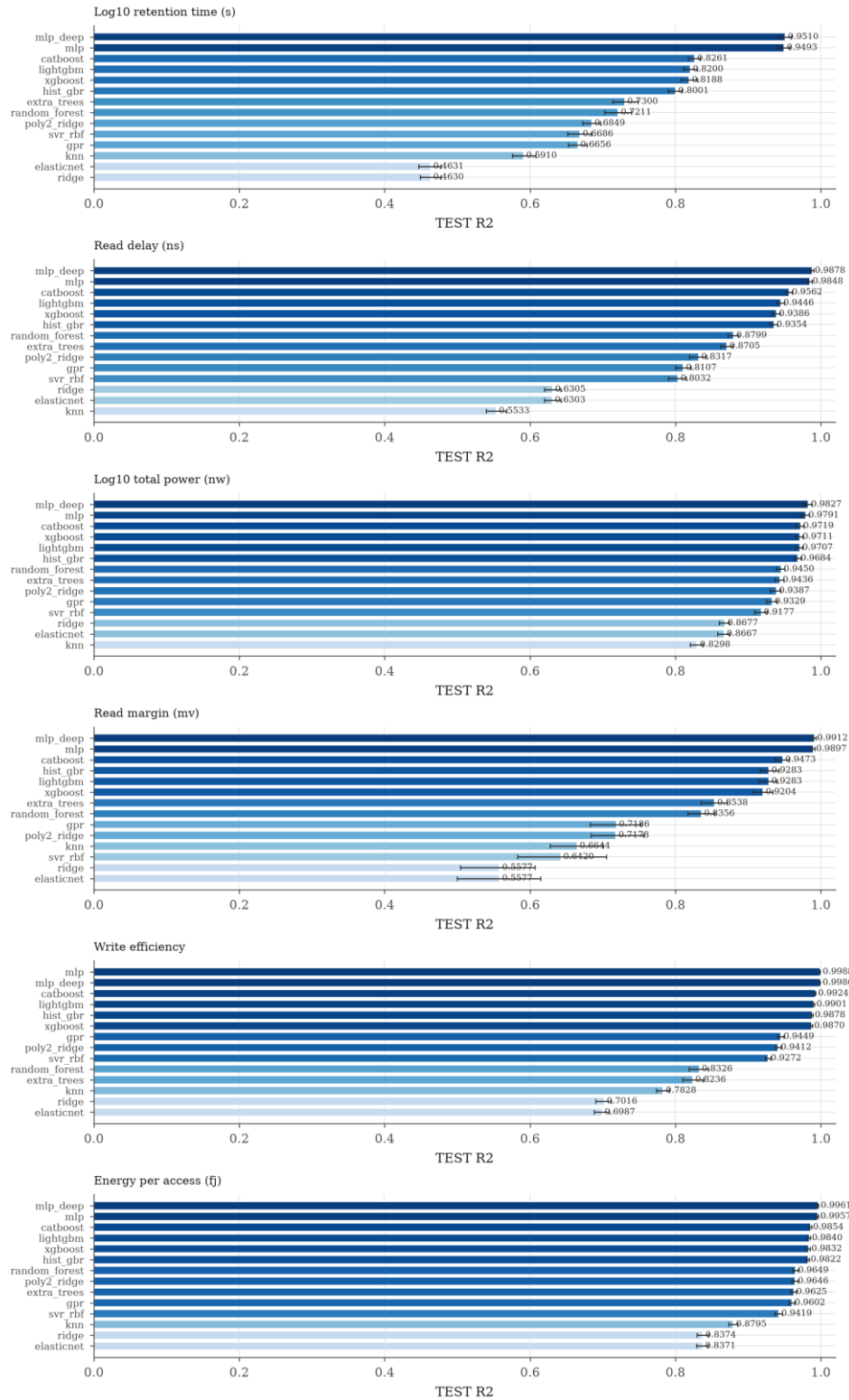


Figure 5. Surrogate model comparison per response.

Digital-twin predictions against SPICE ground truth (test split)

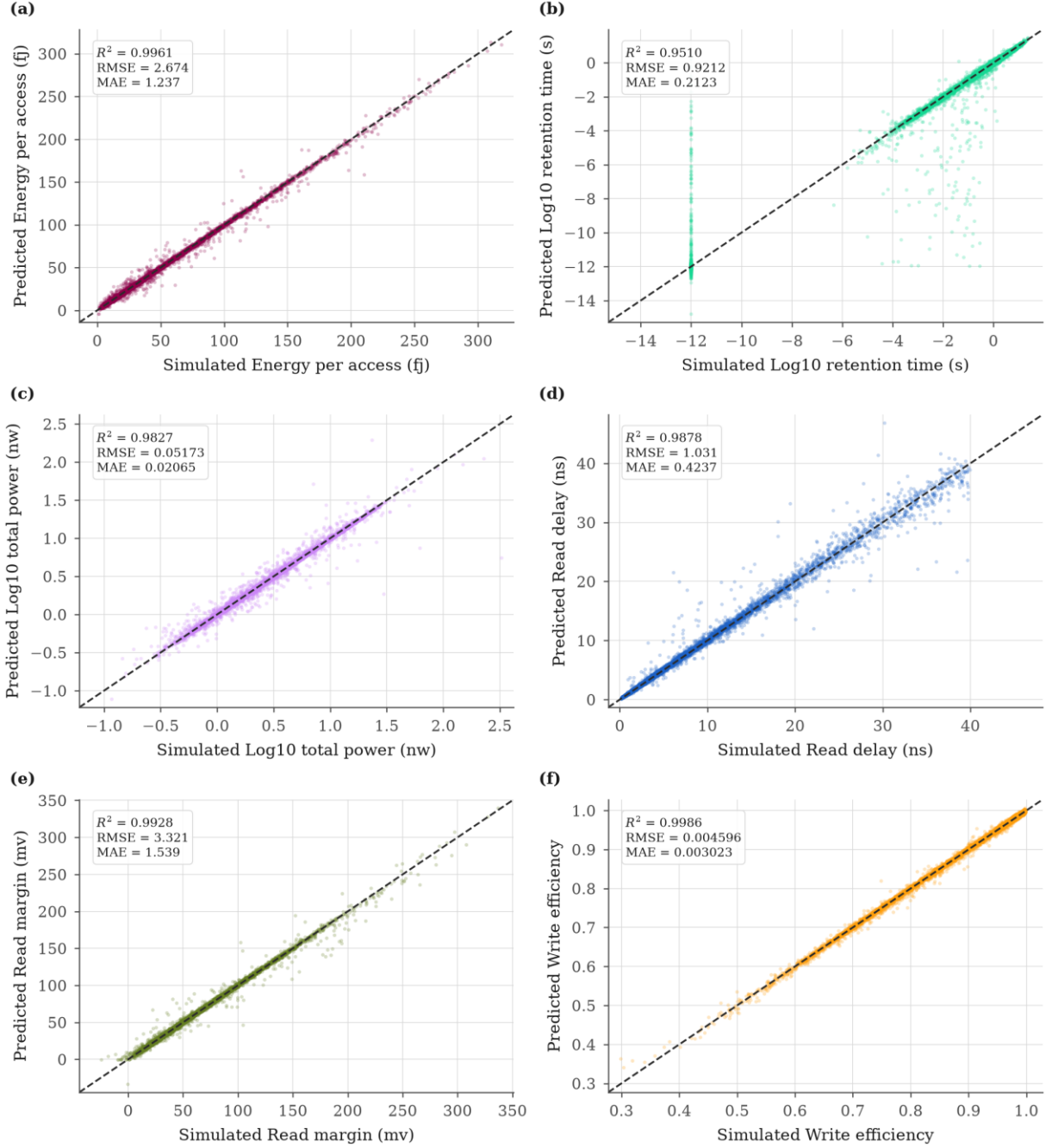


Figure 6. Twin predictions against SPICE ground truth (test split).

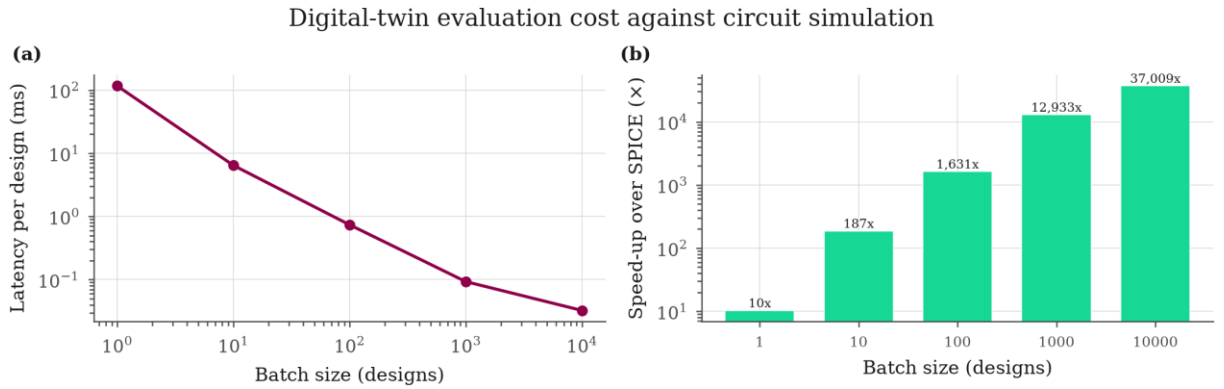
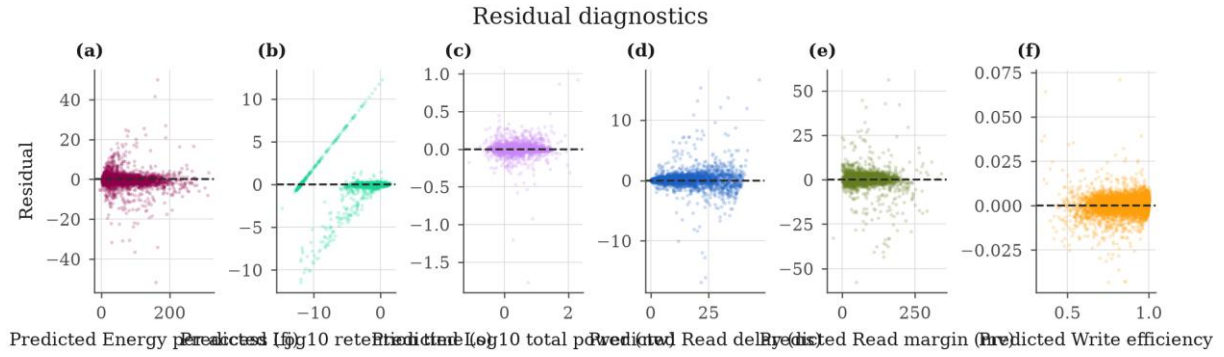


Table 5. Full surrogate comparison.

target	model	family	n_com pare	n_tr ain	cvr2m ean	cvr2s td	test _r2	test_r mse	test_ mae	test_ mape	fitti mes	predictusper _sample
energyperacc ess_fj	mlp_deep	neural	15000	424 91	0.989 6	0.002 397	0.99 61	2.674	1.237	4.853	31.3 4	2.165
energyperacc ess_fj	mlp	neural	15000	424 91	0.989 0	0.001 231	0.99 57	2.834	1.340	5.731	45.7 4	2.783
energyperacc ess_fj	catboost	boosti ng	15000	424 91	0.978 3	0.001 605	0.98 54	5.197	2.723	9.891	6.18 4	2.185
energyperacc ess_fj	lightgbm	boosti ng	15000	424 91	0.974 5	0.002 647	0.98 40	5.451	2.769	9.904	1.49 6	5.291
energyperacc ess_fj	xgboost	boosti ng	15000	424 91	0.972 5	0.003 525	0.98 32	5.576	2.663	9.468	17.4 0	22.10
energyperacc ess_fj	hist_gbr	ense mble	15000	424 91	0.973 8	0.002 531	0.98 22	5.747	3.135	11.21	4.71 0	50.43
energyperacc ess_fj	random_ forest	ense mble	15000	424 91	0.942 6	0.002 066	0.96 49	8.071	4.125	13.75	7.14 8	22.22
energyperacc ess_fj	poly2_rid ge	linear	15000	424 91	0.961 1	0.003 995	0.96 46	8.098	4.732	19.88	0.35 78	3.625
energyperacc ess_fj	extra_tre es	ense mble	15000	424 91	0.940 1	0.002 139	0.96 25	8.331	4.404	14.72	3.16 4	19.97
energyperacc ess_fj	gpr	kernel	15000	200 0	0.955 7	0.002 277	0.96 02	8.593	5.239	20.98	27.1 4	89.03
energyperacc ess_fj	svr_rbf	kernel	15000	500 0	0.941 7	0.003 392	0.94 19	10.38	5.438	19.77	1.58 0	250.8

energyperacc ess_fj	knn	neigh bour	15000	424 91	0.852 6	0.004 291	0.87 95	14.95	9.556	37.58	0.08 240	119.1
energyperacc ess_fj	ridge	linear	15000	424 91	0.835 8	0.006 533	0.83 74	17.36	12.06	56.32	0.08 970	0.9928
energyperacc ess_fj	elasticne t	linear	15000	424 91	0.835 7	0.006 336	0.83 71	17.37	12.07	56.37	15.6 5	1.093
log10retentio ntime_s	mlp_dee p	neural	15000	424 91	0.911 4	0.010 97	0.95 10	0.921 2	0.212 3	46.75	49.8 5	2.363
log10retentio ntime_s	mlp	neural	15000	424 91	0.905 9	0.011 86	0.94 93	0.937 0	0.298 8	88.48	19.2 3	2.406
log10retentio ntime_s	catboost	boosti ng	15000	424 91	0.797 2	0.008 134	0.82 61	1.735	1.016	415.5	6.06 9	1.976
log10retentio ntime_s	lightgbm	boosti ng	15000	424 91	0.784 9	0.006 946	0.82 00	1.765	0.957 7	346.0	1.59 5	5.624
log10retentio ntime_s	xgboost	boosti ng	15000	424 91	0.778 5	0.010 49	0.81 88	1.771	0.934 7	340.0	16.9 0	21.51
log10retentio ntime_s	hist_gbr	ense mble	15000	424 91	0.775 6	0.008 489	0.80 01	1.860	1.057	427.9	2.42 0	29.09
log10retentio ntime_s	extra_tre es	ense mble	15000	424 91	0.693 3	0.007 082	0.73 00	2.162	1.097	367.7	3.09 8	17.68
log10retentio ntime_s	random_ forest	ense mble	15000	424 91	0.680 2	0.010 42	0.72 11	2.197	1.090	399.6	8.08 8	13.02
log10retentio ntime_s	poly2_rid ge	linear	15000	424 91	0.683 7	0.004 772	0.68 49	2.336	1.693	888.7	0.59 85	4.384
log10retentio ntime_s	svr_rbf	kernel	15000	500 0	0.661 5	0.015 27	0.66 86	2.395	1.402	581.1	5.83 0	399.1
log10retentio ntime_s	gpr	kernel	15000	200 0	0.671 6	0.010 09	0.66 56	2.406	1.708	878.6	53.4 1	80.99
log10retentio ntime_s	knn	neigh bour	15000	424 91	0.538 8	0.012 62	0.59 10	2.661	1.538	568.2	0.08 190	123.2
log10retentio ntime_s	elasticne t	linear	15000	424 91	0.462 9	0.008 114	0.46 31	3.049	2.404	1252	10.5 9	1.562
log10retentio ntime_s	ridge	linear	15000	424 91	0.463 0	0.008 211	0.46 30	3.049	2.403	1251	0.09 520	1.018
log10totalpo wer_nw	mlp_dee p	neural	15000	424 91	0.973 6	0.002 399	0.98 27	0.051 73	0.020 65	17.02	53.7 2	3.662
log10totalpo wer_nw	mlp	neural	15000	424 91	0.963 7	0.002 229	0.97 91	0.056 78	0.026 59	25.50	13.3 2	1.522
log10totalpo wer_nw	catboost	boosti ng	15000	424 91	0.970 2	0.002 310	0.97 19	0.065 86	0.038 14	37.11	6.39 7	2.177
log10totalpo wer_nw	xgboost	boosti ng	15000	424 91	0.963 8	0.002 412	0.97 11	0.066 78	0.038 05	36.45	14.4 7	21.99
log10totalpo wer_nw	lightgbm	boosti ng	15000	424 91	0.965 8	0.002 232	0.97 07	0.067 19	0.039 36	38.57	2.47 1	7.269
log10totalpo wer_nw	hist_gbr	ense mble	15000	424 91	0.964 0	0.001 720	0.96 84	0.069 85	0.042 59	41.68	3.56 7	32.45
log10totalpo wer_nw	random_ forest	ense mble	15000	424 91	0.928 0	0.002 960	0.94 50	0.092 09	0.058 06	61.11	7.01 1	18.96
log10totalpo wer_nw	extra_tre es	ense mble	15000	424 91	0.924 3	0.004 048	0.94 36	0.093 26	0.060 36	65.00	3.81 3	20.65
log10totalpo wer_nw	poly2_rid ge	linear	15000	424 91	0.940 4	0.003 131	0.93 87	0.097 27	0.066 61	53.90	0.37 59	3.514
log10totalpo	gpr	kernel	15000	200	0.935	0.003	0.93	0.101	0.068	56.30	43.7	136.6

wer_nw				0	0	826	29	8	06		9	
log10totalpo wer_nw	svr_rbf	kernel	15000	500 0	0.927 4	0.002 894	0.91 77	0.112 7	0.079 15	66.56	2.83 2	833.3
log10totalpo wer_nw	ridge	linear	15000	424 91	0.864 5	0.002 016	0.86 77	0.142 9	0.107 4	93.88	0.13 06	1.167
log10totalpo wer_nw	elasticne t	linear	15000	424 91	0.863 5	0.002 125	0.86 67	0.143 4	0.108 0	95.44	2.13 5	0.9677
log10totalpo wer_nw	knn	neigh bour	15000	424 91	0.790 6	0.002 597	0.82 98	0.162 1	0.115 9	131.7	0.08 870	120.7
readdelayns	mlp_dee p	neural	15000	333 74	0.971 1	0.004 094	0.98 78	1.031	0.423 7	6.293	29.4 4	2.390
readdelayns	mlp	neural	15000	333 74	0.967 9	0.003 776	0.98 48	1.150	0.513 5	10.50	11.7 8	1.693
readdelayns	catboost	boosti ng	15000	333 74	0.936 6	0.002 128	0.95 62	1.952	1.052	17.98	6.03 2	2.390
readdelayns	lightgbm	boosti ng	15000	333 74	0.918 9	0.004 328	0.94 46	2.195	1.137	16.85	1.97 8	6.740
readdelayns	xgboost	boosti ng	15000	333 74	0.906 3	0.003 089	0.93 86	2.311	1.167	15.39	16.2 0	20.61
readdelayns	hist_gbr	ense mble	15000	333 74	0.919 7	0.002 455	0.93 54	2.372	1.328	22.73	4.51 7	53.75
readdelayns	random_ forest	ense mble	15000	333 74	0.849 5	0.004 194	0.87 99	3.233	1.766	21.50	5.35 9	24.07
readdelayns	extra_tre es	ense mble	15000	333 74	0.845 9	0.002 685	0.87 05	3.356	1.840	23.42	2.51 1	21.45
readdelayns	poly2_rid ge	linear	15000	333 74	0.829 4	0.004 340	0.83 17	3.827	2.525	84.66	0.43 78	5.099
readdelayns	gpr	kernel	15000	200 0	0.799 0	0.007 361	0.81 07	4.059	2.788	82.67	52.0 7	136.7
readdelayns	svr_rbf	kernel	15000	500 0	0.810 2	0.005 736	0.80 32	4.138	2.392	51.82	4.64 8	464.6
readdelayns	ridge	linear	15000	333 74	0.632 7	0.005 134	0.63 05	5.670	4.256	161.6	0.07 190	1.203
readdelayns	elasticne t	linear	15000	333 74	0.632 7	0.005 058	0.63 03	5.672	4.255	161.8	21.8 9	2.052
readdelayns	knn	neigh bour	15000	333 74	0.522 6	0.007 825	0.55 33	6.234	4.076	74.97	0.13 32	205.7
readmarginm v	mlp_dee p	neural	15000	424 91	0.951 0	0.012 16	0.99 12	3.675	1.132	4.439	57.4 9	2.367
readmarginm v	mlp	neural	15000	424 91	0.961 4	0.004 524	0.98 97	3.978	1.659	6.609	36.4 9	2.261
readmarginm v	catboost	boosti ng	15000	424 91	0.933 6	0.004 673	0.94 73	8.983	3.386	17.62	6.55 9	2.569
readmarginm v	hist_gbr	ense mble	15000	424 91	0.904 6	0.008 269	0.92 83	10.47	4.057	18.63	4.97 8	53.03
readmarginm v	lightgbm	boosti ng	15000	424 91	0.891 6	0.011 84	0.92 83	10.48	3.648	17.72	1.72 6	5.991
readmarginm v	xgboost	boosti ng	15000	424 91	0.883 9	0.014 71	0.92 04	11.04	3.967	19.11	17.0 5	21.78
readmarginm v	extra_tre es	ense mble	15000	424 91	0.835 7	0.008 539	0.85 38	14.96	7.094	38.52	3.43 8	22.42
readmarginm v	random_ forest	ense mble	15000	424 91	0.808 0	0.013 71	0.83 56	15.86	7.500	39.36	6.98 8	17.41

readmarginmv	gpr	kernel	15000	2000	0.7372	0.01299	0.7186	20.75	10.57	39.31	58.02	126.2
readmarginmv	poly2_ridge	linear	15000	42491	0.7473	0.01231	0.7178	20.78	9.720	34.72	0.3419	3.575
readmarginmv	knn	neighbour	15000	42491	0.6497	0.01848	0.6644	22.66	11.61	42.97	0.08430	123.5
readmarginmv	svr_rbf	kernel	15000	5000	0.6926	0.02552	0.6420	23.40	6.339	17.29	2.409	450.3
readmarginmv	ridge	linear	15000	42491	0.6106	0.02354	0.5577	26.01	12.60	40.29	0.2018	1.751
readmarginmv	elasticnet	linear	15000	42491	0.6104	0.02377	0.5577	26.02	12.52	40.14	23.02	1.013
write_efficiency	mlp	neural	15000	42491	0.9880	0.004424	0.9988	0.004309	0.002680	0.3280	40.56	1.779
write_efficiency	mlp_deep	neural	15000	42491	0.9931	0.002262	0.9986	0.004596	0.003023	0.3674	80.50	2.752
write_efficiency	catboost	boosting	15000	42491	0.9846	0.001761	0.9924	0.01068	0.006261	0.7495	5.534	1.594
write_efficiency	lightgbm	boosting	15000	42491	0.9788	0.002283	0.9901	0.01217	0.006683	0.8245	1.402	5.353
write_efficiency	hist_gbr	ensemble	15000	42491	0.9780	0.002281	0.9878	0.01350	0.007959	0.9693	2.835	32.42
write_efficiency	xgboost	boosting	15000	42491	0.9707	0.003061	0.9870	0.01396	0.007417	0.9326	10.26	9.803
write_efficiency	gpr	kernel	15000	2000	0.9405	0.003055	0.9449	0.02871	0.02047	2.419	30.50	72.07
write_efficiency	poly2_ridge	linear	15000	42491	0.9330	0.003018	0.9412	0.02967	0.02172	2.574	0.3446	3.663
write_efficiency	svr_rbf	kernel	15000	5000	0.9268	0.004381	0.9272	0.03299	0.02446	2.911	0.3353	74.83
write_efficiency	random_forest	ensemble	15000	42491	0.8115	0.007213	0.8326	0.05004	0.02968	3.796	7.803	17.87
write_efficiency	extra_trees	ensemble	15000	42491	0.8035	0.008971	0.8236	0.05137	0.03042	3.906	3.020	18.00
write_efficiency	knn	neighbour	15000	42491	0.7358	0.005847	0.7828	0.05700	0.03677	4.861	0.08340	119.1
write_efficiency	ridge	linear	15000	42491	0.6884	0.002428	0.7016	0.06681	0.05356	6.432	0.09010	1.065
write_efficiency	elasticnet	linear	15000	42491	0.6850	0.002234	0.6987	0.06713	0.05354	6.455	0.8738	1.127

cv* are 5-fold cross-validated means over *ncompare* rows of the train+val partition; the model is then refitted on all *ntrain* rows and scored once on the held-out test partition. Kernel methods carry a further per-model cap (see the notes column)._

Table 6. Best surrogate per response with bootstrap CIs.

response	best model	family	n train	n test	test r2	test rmse	test mae	test map_e	test max_error	test r2 95% CI	test rmse 95% CI	test mae 95% CI
log10retentiontime_s	mlp_deep	neural	42491	7496	0.9510	0.9212	0.2123	46.75	12.24	[0.9413,	[0.8323,	[0.1927,

										0.9594]	1.0117]	0.2322]
readdelayns	mlp_de ep	neur al	3337 4	591 5	0.987 8	1.031	0.4237	6.293	16.84	[0.985 2, 0.9901]	[0.903 7, 1.1546]	[0.400 2, 0.4489]
log10totalpower _nw	mlp_de ep	neur al	4249 1	749 6	0.982 7	0.0517 3	0.0206 5	17.02	1.767	[0.975 0, 0.9875]	[0.043 1, 0.0606]	[0.019 7, 0.0218]
readmarginmv	mlp	neur al	4249 1	749 6	0.992 8	3.321	1.539	6.071	57.69	[0.991 5, 0.9943]	[2.943 0, 3.6440]	[1.471 9, 1.6086]
write_efficiency	mlp_de ep	neur al	4249 1	749 6	0.998 6	0.0045 96	0.0030 23	0.367 4	0.07089	[0.998 4, 0.9987]	[0.004 3, 0.0049]	[0.002 9, 0.0031]
energyperaccess_ fj	mlp_de ep	neur al	4249 1	749 6	0.996 1	2.674	1.237	4.853	51.67	[0.995 4, 0.9968]	[2.404 5, 2.9008]	[1.182 7, 1.2955]

Table 7. Digital-twin latency and speed-up.

batch_size	total_ms	msperdesign	designspersecond	spicemsper_design	speedupvsspice
1	116.3	116.3	8.600	1202	10.30
10	64.25	6.425	155.6	1202	187.1
100	73.71	0.7371	1357	1202	1631
1000	92.95	0.09295	10758	1202	12933
10000	324.8	0.03248	30786	1202	37009

The SPICE reference is the mean simulation time actually recorded while generating the training dataset.

7. Multi-objective optimisation

Six optimisers were each run 7 times under an identical budget of 4000 surrogate evaluations. Ranked by median hypervolume, **NSGA-II** leads (2851.5).

Algorithm	Median HV	Median IGD+	Median spacing	Median solutions	Median runtime (s)
NSGA-II	2851.5	8.8681	3.375	100	4.4
NSGA-III	2700.3	9.5737	6.094	40	4.5
Bayesian optimisation (multi-objective TPE)	2461.8	11.986	2.611	245	551.4
Differential evolution	905.67	31.3	8.877	12	4.6
Particle-swarm optimisation	891.45	33.442	8.088	13	4.9
MOEA/D	434.56	0.1903	0.2041	99	249.2

The Friedman/Nemenyi analysis on hypervolume gives a critical difference of 2.850; algorithms whose mean ranks differ by less than that are statistically indistinguishable at $\alpha = 0.05$ and are joined in the critical-difference diagram.

SPICE verification of the selected designs. The Pareto designs were returned to NGSpice and re-simulated at every operating condition. Median relative error between twin prediction and simulation: *retentiontimes* 46.33 %, *readdelays* 30.82 %, *totalpowernw* 10.31 %, *readmarginmv* 3.04 %.

12 representative designs were selected (knee point, weighted compromise and per-objective extremes); they are tabulated below and carried into the robustness study.

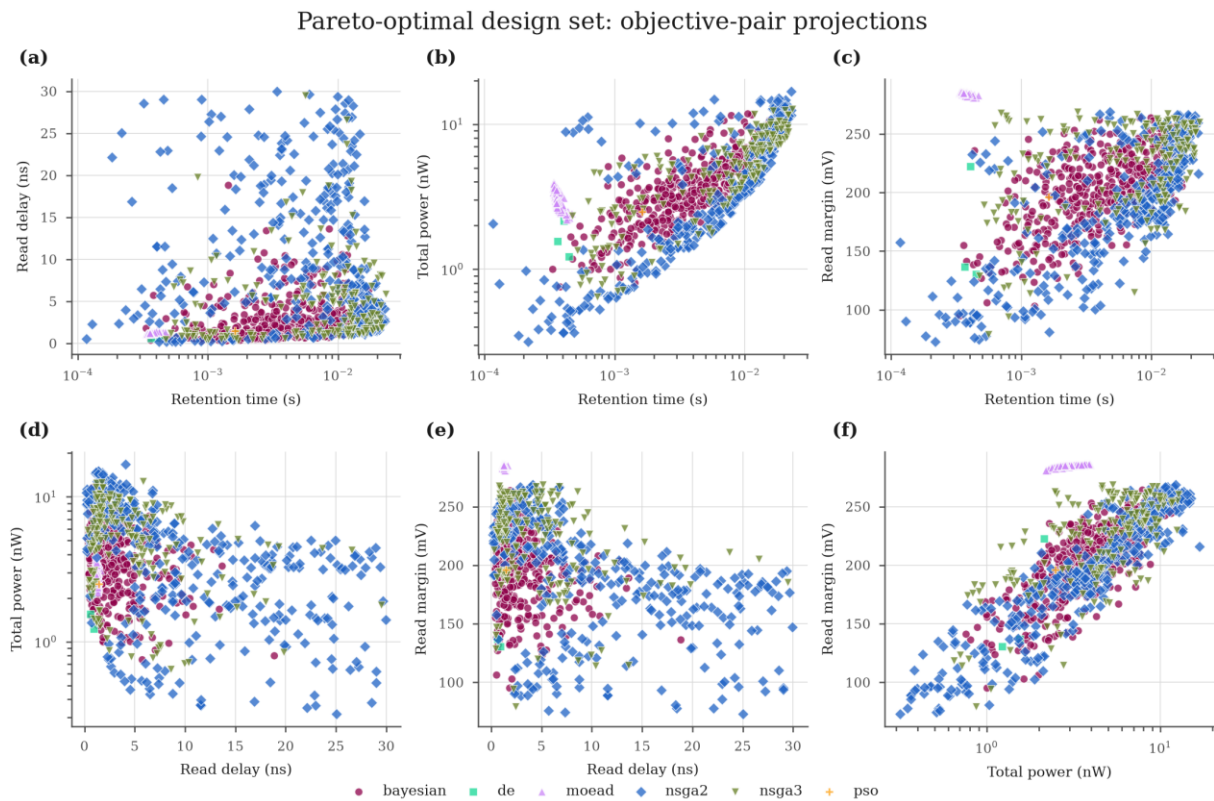


Figure 9. Pareto-optimal design set, objective-pair projections.

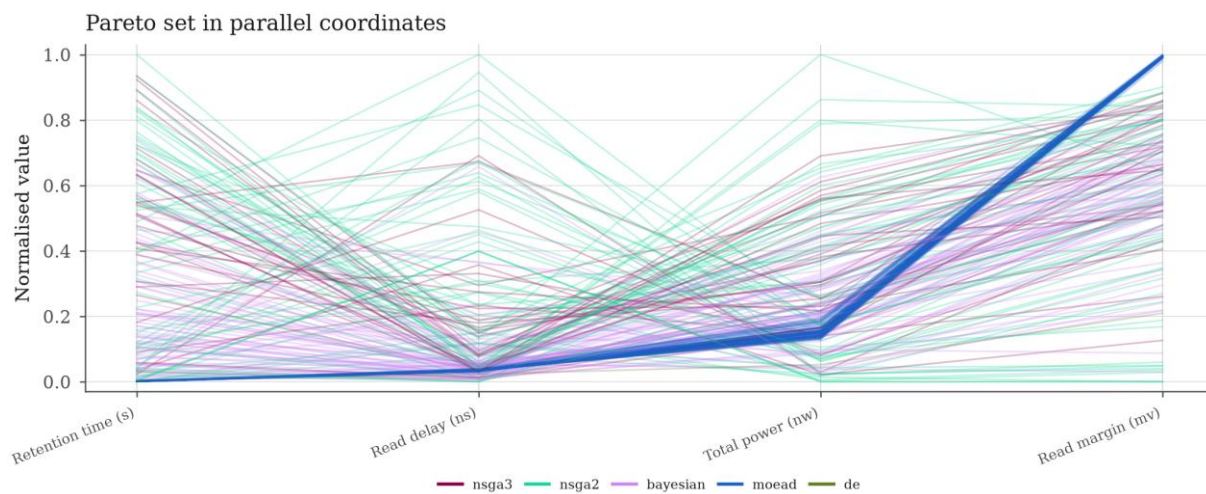


Figure 10. Pareto set in parallel coordinates.

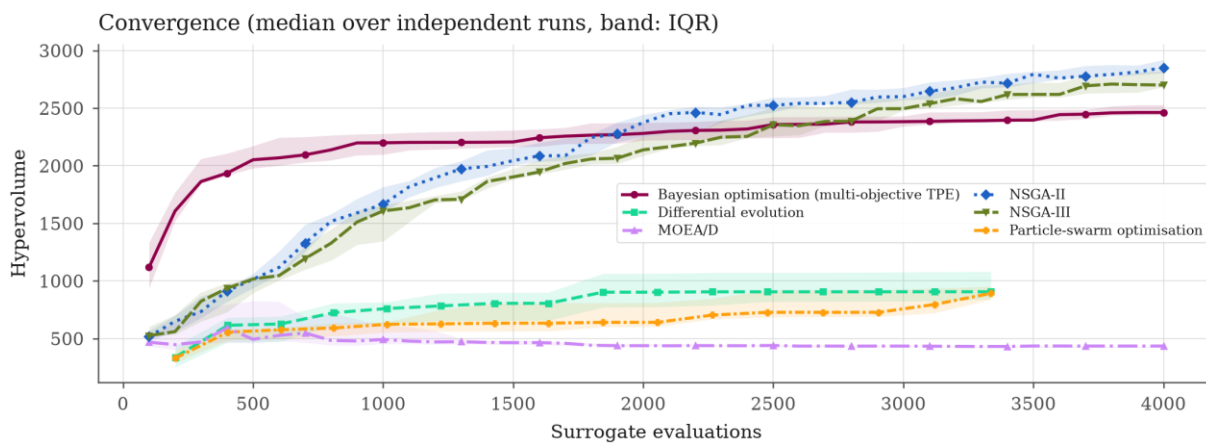


Figure 11. Hypervolume convergence (median over runs, IQR band).

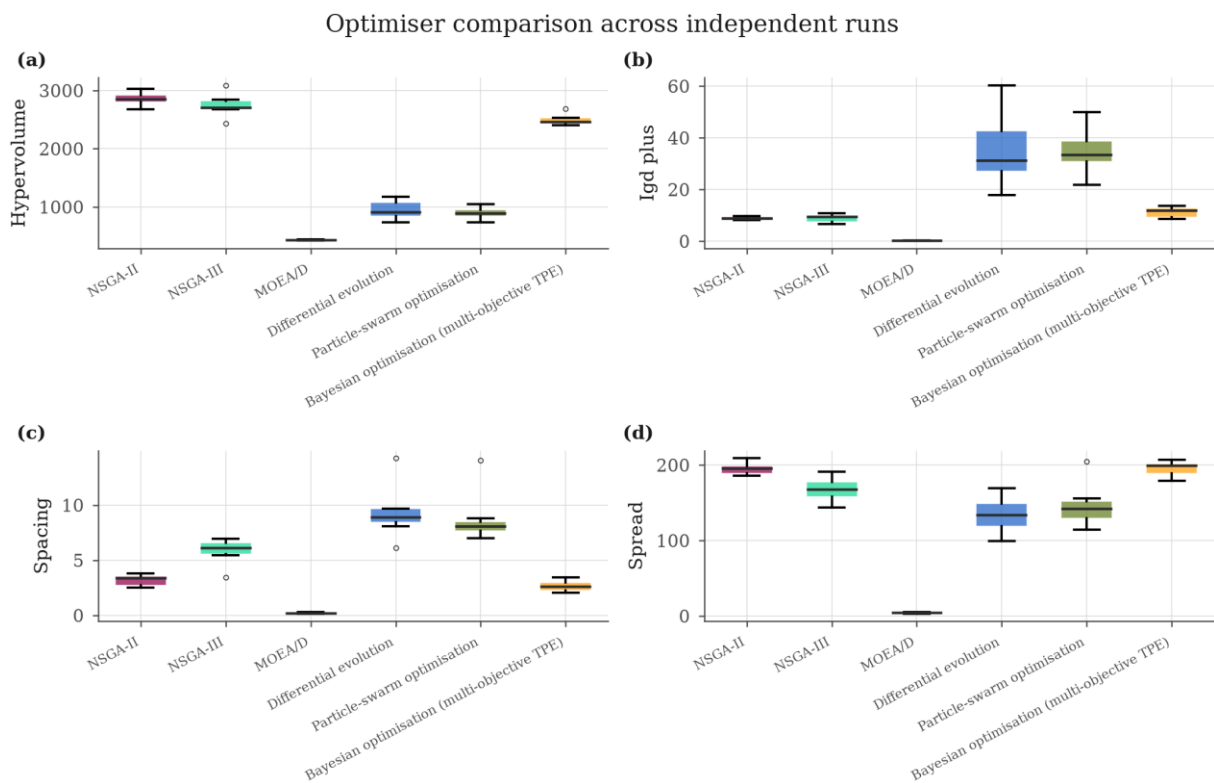


Figure 12. Optimiser comparison across independent runs.

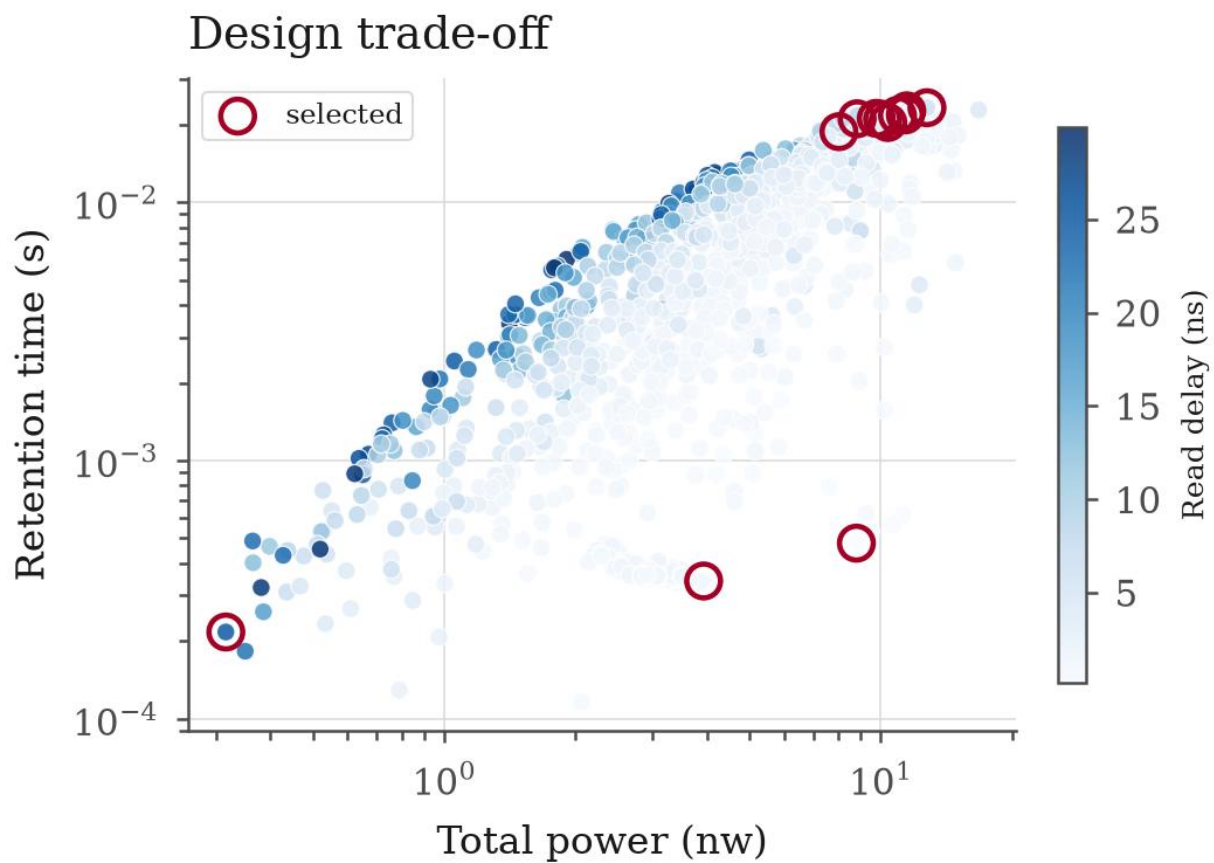
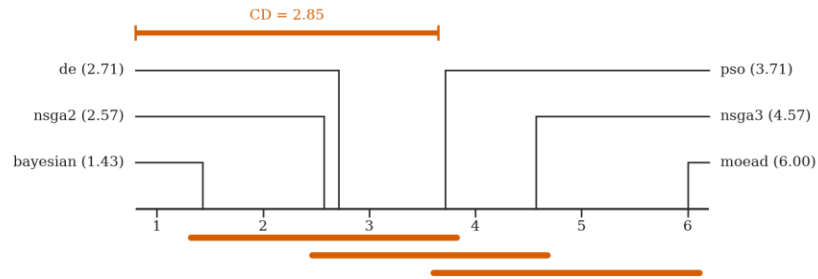


Figure 13. Design trade-off with the selected designs highlighted.

Critical-difference diagram: optimizer_epsilon



mean Friedman rank (lower is better); methods joined by a bar are not significantly different

Figure 14. Critical-difference diagram (Nemenyi post-hoc).

Table 8. Optimiser comparison.

algorithm	label	runs	hv_median	hv_mean	hv_std	igdplusmedian	spacing_median	spread_median	nsolutionsmedian	runtimemedian
nsga2	NSGA-II	7	2851	2857	118.0	8.868	3.375	195.2	100	4.351
nsga3	NSGA-III	7	2700	2748	198.2	9.574	6.094	167.9	40	4.467
bayesian	Bayesian optimisation (multi-objective TPE)	7	2462	2498	93.49	11.99	2.611	198.9	245	551.4
de	Differential evolution	7	905.7	950.5	163.2	31.30	8.877	134.0	12	4.564
pso	Particle-swarm optimisation	7	891.4	892.2	109.0	33.44	8.088	142.4	13	4.878
moead	MOEA/D	7	434.6	429.4	7.685	0.1903	0.2041	4.208	99	249.2

Budget 4000 surrogate evaluations per run, 7 runs per algorithm. Hypervolume and IGD+ use a reference front built from the non-dominated union of every run.

Table 9. Representative Pareto-optimal designs.

selection	found_by	vd	vwl_best	vw_low	vblp_ratio	cs	cbl_ratio	w_acc	la_cc	ws_an	wsa_ratio	we_ratio	twri_tepulse	tsen_sedelay	retentiontimes	read_delays	total_powernw	read_marginmv
knee	nsga3	1.299	0.8995	-0.008576	0.3569	3.527e-14	2.074	6.015e-8	5.846e-8	1.415e-7	2.474	1.234	8.694e-9	8.593e-9	0.01879	4.997	8.018	245.9

weight ed	nsg a3	1. 22 7	0.9 809	- 0.0 35 07	0.35 00	3. 98 7e - 14	2.0 04	6. 53 4e -8	8. 10 4e -8	2. 01 5e -7	2.9 87	3.8 02	9.6 02e -9	1.94 9e-9	0.021 08	2.99 3	8.82 4	230.6
bestret entiont ime_s	nsg a3	1. 28 7	1.1 21	- 0.0 23 29	0.35 04	3. 99 6e - 14	2.0 71	6. 24 0e -8	6. 56 5e -8	1. 08 3e -7	2.7 24	3.8 23	9.5 11e -9	8.50 3e-9	0.023 26	5.84 9	12.7 5	260.3
bestre addela y_ns	nsg a2	1. 29 4	1.0 88	- 0.1 15 9	0.35 65	5. 16 2e - 15	2.0 37	2. 09 6e -7	6. 41 6e -8	4. 79 0e -7	1.8 57	3.1 48	6.3 20e -9	5.98 1e-9	4.789 e-4	0.19 95	8.79 0	225.8
besttot alpowe r_nw	nsg a2	0. 70 90	0.6 362	- 0.0 56 50	0.63 35	5. 30 8e - 15	2.0 05	8. 93 6e -8	8. 23 2e -8	1. 44 2e -7	2.3 13	1.5 31	9.6 67e -9	3.17 6e-9	2.168 e-4	25.0 4	0.31 61	72.80
bestre admar gin_mv	mo ea d	1. 30 0	0.9 965	- 0.3 99 8	0.35 00	5. 00 6e - 15	2.0 00	2. 20 0e -7	1. 20 0e -7	9. 00 0e -8	1.0 00	2.2 08	5.0 71e -9	1.28 7e-9	3.418 e-4	1.17 1	3.92 8	286.2
compr omise	nsg a2	1. 28 5	0.9 786	- 0.0 59 26	0.35 14	3. 93 9e - 14	2.0 05	6. 54 4e -8	6. 23 7e -8	1. 81 9e -7	2.7 97	3.9 42	9.5 69e -9	9.59 4e-9	0.020 97	2.94 0	10.0 9	261.5
compr omise	nsg a2	1. 30 0	1.0 53	- 0.0 03 78 7	0.35 31	3. 96 7e - 14	2.0 04	6. 27 8e -8	7. 49 4e -8	1. 76 4e -7	2.9 76	3.6 41	9.2 98e -9	6.10 5e-9	0.022 63	2.89 9	11.4 7	259.5
compr omise	nsg a3	1. 24 7	1.0 23	- 0.0 25 45	0.35 36	3. 95 3e - 14	2.0 02	6. 49 4e -8	8. 82 6e -8	2. 31 8e -7	2.9 40	3.8 85	9.6 12e -9	2.47 7e-9	0.021 16	2.46 4	9.76 4	237.0
compr omise	nsg a2	1. 27 0	1.0 59	- 0.0 02 68 6	0.35 31	3. 96 7e - 14	2.0 46	6. 26 0e -8	6. 86 9e -8	1. 76 6e -7	2.9 78	3.6 41	9.6 30e -9	6.10 5e-9	0.021 72	3.14 1	10.9 2	256.5
compr omise	nsg a2	1. 28 0	1.0 82	- 0.0 38 65	0.35 25	3. 99 8e - 14	2.0 02	6. 07 7e -8	6. 11 4e -8	2. 30 1e -7	2.7 90	2.9 98	9.2 00e -9	8.91 4e-9	0.021 83	2.55 3	11.3 8	259.1
compr omise	nsg a3	1. 29 9	1.0 19	- 0.0 24 42	0.35 48	3. 79 0e -	2.0 41	6. 03 8e -8	6. 71 2e -8	2. 49 4e -7	1.9 48	2.9 11	9.2 93e -9	5.80 7e-9	0.020 39	2.58 3	10.3 7	258.0

					14												
--	--	--	--	--	----	--	--	--	--	--	--	--	--	--	--	--	--

Objective values are worst case over the configured PVT conditions.

Table 10. Twin prediction versus SPICE for the selected designs.

sel ecti on	f o u n d _ b y	te m p e r a t u r e _ c	c o n f i g u r e	rete ntio ntim es_p redict ed	rete ntio ntim es_si mul tated	rea ddel ayn s_pr edic ted	rea ddel ayn s_si mul tate d	total pow ern w_p redi cted	total pow ern w_si mul tated	read mar gin mv_ pred icte d	read mar gin mv_ simu late d	writ eeffi cien cys mul tated	re ad_ s uc ce ss	w rit e_ su cc es s	rete ntio ntim esrel error _pct	read delays r_ eler ror_ pct	total pow ern wrel erro r_pc t	read mar ginm vrel ror_ pct
kne e	n s g a 3	27	T T	36.2 1	24.5 8	2.12 8	1.76 6	6.83 0	6.78 8	259. 6	267. 8	0.99 79	ye s	ye s	47.2 9	20.5 1	0.62 25	3.06 6
kne e	n s g a 3	85	S S	1.15 1	0.81 01	3.79 2	2.86 1	6.22 5	6.97 4	253. 0	264. 3	0.98 93	ye s	ye s	42.0 7	32.5 6	10.7 5	4.26 0
kne e	n s g a 3	12 5	S S	0.01 879	0.01 220	4.99 7	3.80 0	6.48 8	7.23 2	245. 9	259. 4	0.97 72	ye s	ye s	54.1 0	31.5 0	10.2 8	5.20 7
kne e	n s g a 3	85	F F	1.04 4	0.81 01	2.68 2	2.86 1	8.01 8	6.97 4	260. 2	264. 3	0.98 93	ye s	ye s	28.8 3	6.25 5	14.9 6	1.53 1
wei ght ed	n s g a 3	27	T T	46.4 7	29.2 3	1.21 1	1.01 3	7.35 6	7.08 4	255. 7	260. 1	0.99 88	ye s	ye s	58.9 9	19.5 7	3.82 9	1.71 3
wei ght ed	n s g a 3	85	S S	1.30 0	0.85 70	2.24 3	1.59 8	6.97 4	7.36 7	242. 5	247. 0	0.99 27	ye s	ye s	51.6 5	40.4 1	5.34 0	1.82 3
wei ght ed	n s g a 3	12 5	S S	0.02 108	0.01 238	2.99 3	2.11 6	7.35 2	7.84 5	230. 6	224. 1	0.97 95	ye s	ye s	70.2 8	41.4 3	6.27 9	2.89 9
wei ght ed	n s g a 3	85	F F	1.21 8	0.85 70	1.59 8	1.59 8	8.82 4	7.36 7	249. 0	247. 0	0.99 27	ye s	ye s	42.0 8	0.03 369	19.7 8	0.79 01

bes tret ent ion tim e_s	n s g a 3	27	T T	44.0 2	28.4 8	2.39 4	1.88 4	10.4 1	9.43 2	261. 3	270. 1	0.99 89	ye s	ye s	54.5 6	27.1 2	10.3 4	3.27 6
bes tret ent ion tim e_s	n s g a 3	85	S S	1.27 5	0.89 77	4.41 4	2.98 4	8.85 6	9.80 8	261. 2	269. 9	0.99 82	ye s	ye s	42.0 2	47.9 3	9.70 4	3.21 5
bes tret ent ion tim e_s	n s g a 3	12 5	S S	0.02 326	0.01 383	5.84 9	3.90 9	9.52 1	10.2 5	260. 3	268. 3	0.99 41	ye s	ye s	68.1 9	49.6 1	7.10 5	3.00 1
bes tret ent ion tim e_s	n s g a 3	85	F F	1.20 2	0.89 77	3.24 1	2.98 4	12.7 5	9.80 8	261. 6	269. 9	0.99 82	ye s	ye s	33.8 6	8.62 4	30.0 5	3.08 9
bes tre add ela y_n s	n s g a 2	27	T T	2.98 1	3.15 4	0.00 866 6	0.11 92	4.93 3	4.45 0	225. 8	239. 7	0.97 11	ye s	ye s	5.46 7	92.7 3	10.8 5	5.80 0
bes tre add ela y_n s	n s g a 2	85	S S	0.03 029	0.03 603	0.16 09	0.18 34	4.27 2	4.97 7	229. 6	241. 1	0.97 05	ye s	ye s	15.9 2	12.3 1	14.1 8	4.77 1
bes tre add ela y_n s	n s g a 2	12 5	S S	4.78 9e-4	5.85 3e-4	0.19 95	0.23 56	4.99 2	5.90 3	230. 6	242. 0	0.97 01	ye s	ye s	18.1 8	15.3 1	15.4 4	4.73 1
bes tre add ela y_n s	n s g a 2	85	F F	0.03 192	0.03 603	0.04 727	0.18 34	8.79 0	4.97 7	228. 0	241. 1	0.97 05	ye s	ye s	11.4 1	74.2 3	76.6 2	5.42 6
bes ttot alp ow er_ nw	n s g a 2	27	T T	0.81 99	1.04 5	9.17 3	10.3 4	0.27 54	0.25 64	79.1 7	82.7 2	0.98 63	ye s	ye s	21.5 3	11.2 5	7.40 2	4.28 5

besttotal power_nw	ns	85	SS	0.01185	0.02169	25.04	10.91	0.2540	0.2777	72.80	82.54	0.9846	yes	yes	45.38	129.4	8.512	11.80
besttotal power_nw	ns	125	SS	2.168e-4	3.136e-4	23.79	11.17	0.2762	0.3488	74.60	82.49	0.9834	yes	yes	30.86	113.1	20.82	9.563
besttotal power_nw	ns	85	FF	0.02051	0.02169	4.950	10.91	0.3161	0.2777	84.88	82.54	0.9846	yes	yes	5.470	54.65	13.84	2.829
bestreadmargin_mv	moead	27	TT	1.988	1.959	0.5659	0.6243	2.707	2.721	286.2	295.5	0.9495	yes	yes	1.506	9.351	0.5139	3.136
bestreadmargin_mv	moead	85	SS	0.01821	0.01834	0.9472	1.006	2.210	2.937	286.2	297.2	0.9489	yes	yes	0.6982	5.867	24.75	3.708
bestreadmargin_mv	moead	125	SS	3.418e-4	3.363e-4	1.171	1.321	2.473	3.277	286.2	298.2	0.9484	yes	yes	1.638	11.36	24.54	4.024
bestreadmargin_mv	moead	85	FF	0.01756	0.01834	0.7144	1.006	3.928	2.937	287.1	297.2	0.9489	yes	yes	4.237	29.00	33.71	3.374
compromise	ns	27	TT	42.05	27.26	1.331	1.023	8.232	7.824	264.8	273.1	0.9990	yes	yes	54.24	30.14	5.221	3.025
compromise	ns	85	SS	1.228	0.8148	2.246	1.613	7.473	8.126	264.6	272.1	0.9964	yes	yes	50.72	39.22	8.036	2.756

mis e	a 2																	
co mp ro mis e	n s g a 2	12 5	S S	0.02 097	0.01 294	2.94 0	2.11 3	8.03 3	8.60 4	261. 5	269. 3	0.98 94	ye s	ye s	62.0 6	39.1 2	6.63 8	2.88 8
co mp ro mis e	n s g a 2	85	F F	1.20 6	0.81 48	1.66 6	1.61 3	10.0 9	8.12 6	265. 2	272. 1	0.99 64	ye s	ye s	48.0 5	3.28 9	24.1 8	2.52 7
co mp ro mis e	n s g a 2	27	T T	48.0 6	28.3 5	1.34 9	1.02 0	9.37 0	8.74 9	266. 5	275. 6	0.99 89	ye s	ye s	69.5 0	32.2 7	7.10 2	3.29 9
co mp ro mis e	n s g a 2	85	S S	1.29 1	0.90 13	2.27 2	1.61 0	8.43 6	9.11 2	266. 1	274. 4	0.99 59	ye s	ye s	43.2 7	41.1 1	7.42 7	3.02 3
co mp ro mis e	n s g a 2	12 5	S S	0.02 263	0.01 362	2.89 9	2.11 4	8.94 9	9.63 2	259. 5	270. 4	0.98 63	ye s	ye s	66.1 3	37.1 0	7.09 6	3.99 8
co mp ro mis e	n s g a 2	85	F F	1.23 7	0.90 13	1.68 3	1.61 0	11.4 7	9.11 2	266. 4	274. 4	0.99 59	ye s	ye s	37.2 9	4.52 7	25.8 3	2.91 5
co mp ro mis e	n s g a 3	27	T T	47.5 1	28.4 5	0.97 84	0.84 06	7.88 7	7.66 5	257. 4	262. 8	0.99 88	ye s	ye s	67.0 1	16.4 0	2.89 7	2.04 9
co mp ro mis e	n s g a 3	85	S S	1.28 8	0.84 67	1.84 5	1.31 7	7.50 6	8.00 4	248. 7	255. 7	0.99 35	ye s	ye s	52.0 9	40.0 8	6.22 2	2.73 7
co mp ro mis e	n s g a 3	12 5	S S	0.02 116	0.01 249	2.46 4	1.74 0	7.91 8	8.56 5	237. 0	238. 3	0.98 02	ye s	ye s	69.4 0	41.6 3	7.55 2	0.54 59
co mp ro mis e	n s g a 3	85	F F	1.15 8	0.84 67	1.36 7	1.31 7	9.76 4	8.00 4	254. 2	255. 7	0.99 35	ye s	ye s	36.7 8	3.77 2	21.9 9	0.58 13
co mp	n s	27	T T	47.0 1	28.6 8	1.39 8	1.10 7	8.95 1	8.37 5	258. 4	265. 7	0.99 90	ye s	ye s	63.8 9	26.3 0	6.88 8	2.73 1

ro mis e	g a 2																	
co mp ro mis e	n s g a 2	85	S S	1.33 0	0.89 04	2.42 8	1.74 2	8.11 4	8.70 9	258. 2	265. 1	0.99 74	ye s	ye s	49.3 9	39.3 9	6.82 7	2.58 0
co mp ro mis e	n s g a 2	12 5	S S	0.02 172	0.01 315	3.14 1	2.27 7	8.59 9	9.18 8	256. 5	262. 6	0.99 11	ye s	ye s	65.1 3	37.9 7	6.41 3	2.29 6
co mp ro mis e	n s g a 2	85	F F	1.22 4	0.89 04	1.75 1	1.74 2	10.9 2	8.70 9	258. 7	265. 1	0.99 74	ye s	ye s	37.5 3	0.54 26	25.3 5	2.41 8

Table 11. *Friedman ranks of the optimisers.*

indicator	group	mean_rank	rank_position	critical_difference
hypervolume	nsga2	1.286	1	2.850
hypervolume	nsga3	1.857	2	2.850
hypervolume	bayesian	2.857	3	2.850
hypervolume	de	4.286	4	2.850
hypervolume	pso	4.714	5	2.850
hypervolume	moead	6	6	2.850
igd	nsga2	1.714	1	2.850
igd	bayesian	2.143	2	2.850
igd	nsga3	2.143	3	2.850
igd	de	4.571	4	2.850
igd	pso	4.714	5	2.850
igd	moead	5.714	6	2.850
igd_plus	moead	1	1	2.850
igd_plus	nsga3	2.571	2	2.850
igd_plus	nsga2	3	3	2.850
igd_plus	bayesian	3.429	4	2.850
igd_plus	de	5.429	5	2.850
igd_plus	pso	5.571	6	2.850
spacing	moead	1	1	2.850
spacing	bayesian	2.143	2	2.850
spacing	nsga2	3	3	2.850
spacing	nsga3	4	4	2.850
spacing	pso	5.286	5	2.850
spacing	de	5.571	6	2.850
spread	bayesian	1.571	1	2.850
spread	nsga2	1.857	2	2.850
spread	nsga3	3.429	3	2.850
spread	pso	3.857	4	2.850
spread	de	4.286	5	2.850
spread	moead	6	6	2.850

epsilon	bayesian	1.429	1	2.850
epsilon	nsga2	2.571	2	2.850
epsilon	de	2.714	3	2.850
epsilon	pso	3.714	4	2.850
epsilon	nsga3	4.571	5	2.850
epsilon	moead	6	6	2.850
n_solutions	bayesian	1	1	2.850
n_solutions	nsga2	2.143	2	2.850
n_solutions	moead	2.857	3	2.850
n_solutions	nsga3	4	4	2.850
n_solutions	pso	5.429	5	2.850
n_solutions	de	5.571	6	2.850

Ranks are computed per run after orienting each indicator so that a smaller value is better.

8. Explainability, robustness and assist techniques

Explainability. Across all modelled responses the highest-attribution design variables are num__vdd, num__vwl_overdrive, num__signal_charge_c, num__charge_share_ratio, num__vwl_boost, num__t_write_pulse. SHAP gives the global ranking and the direction of each effect; ALE is reported alongside partial dependence because the derived design variables are correlated by construction, which is exactly the situation in which partial dependence becomes misleading.

Sensitivity. Sobol' total-effect indices per response:

- read_delay_ns: vdd ($ST = 0.761$, interaction 0.141), cblratio ($ST = 0.172$, interaction 0.106), cs ($ST = 0.108$, interaction 0.075)
- read_margin_mv: cblratio ($ST = 0.510$, interaction 0.145), vwlboost ($ST = 0.317$, interaction 0.097), vblpreratio ($ST = 0.203$, interaction 0.140)
- retention_time_s: cs ($ST = 0.384$, interaction 0.203), vblpreratio ($ST = 0.348$, interaction 0.134), cblratio ($S_T = 0.288$, interaction 0.079)
- total_power_nw: vdd ($ST = 0.652$, interaction 0.291), cs ($ST = 0.344$, interaction 0.156), vwlwlow ($ST = 0.315$, interaction 0.311)

A large $S_T - S_1$ gap identifies variables whose influence is mostly through interaction, which one-variable-at-a-time sweeps cannot reveal.

Robustness. Joint Monte-Carlo yield of the selected designs ranges from 100.00 % to 100.00 % (median 100.00 %) under the configured process sigmas. A random subset of the same Monte-Carlo trials was re-simulated in NGSpice; the largest twin-versus-SPICE yield discrepancy was 0.00 percentage points.

Assist techniques. Median change against the unassisted baseline:

Technique	retentiontimes (%)	readdelayns (%)	readmarginmv (%)	totalpowernw (%)
bitlineprechargeopt	+213.9	-2.1	+24.8	+14.7
combinedlowpower	-1.8	-6.2	+70.9	+90.3
highkcapacitor	-81.2	+41.6	-8.6	+50.5
negative_wordline	-37.0	+0.1	-0.0	-0.0
refreshintervalopt	+0.0	+0.0	+0.0	+0.0
senseampupsize	-0.0	-31.6	-1.1	+12.5

wordline_boost	+346.7	-9.7	+55.9	+6.3
----------------	--------	------	-------	------

The study is paired: every technique is applied to the same base designs and simulated in NGSpice, and the differences are tested with paired Wilcoxon signed-rank tests under Holm correction.

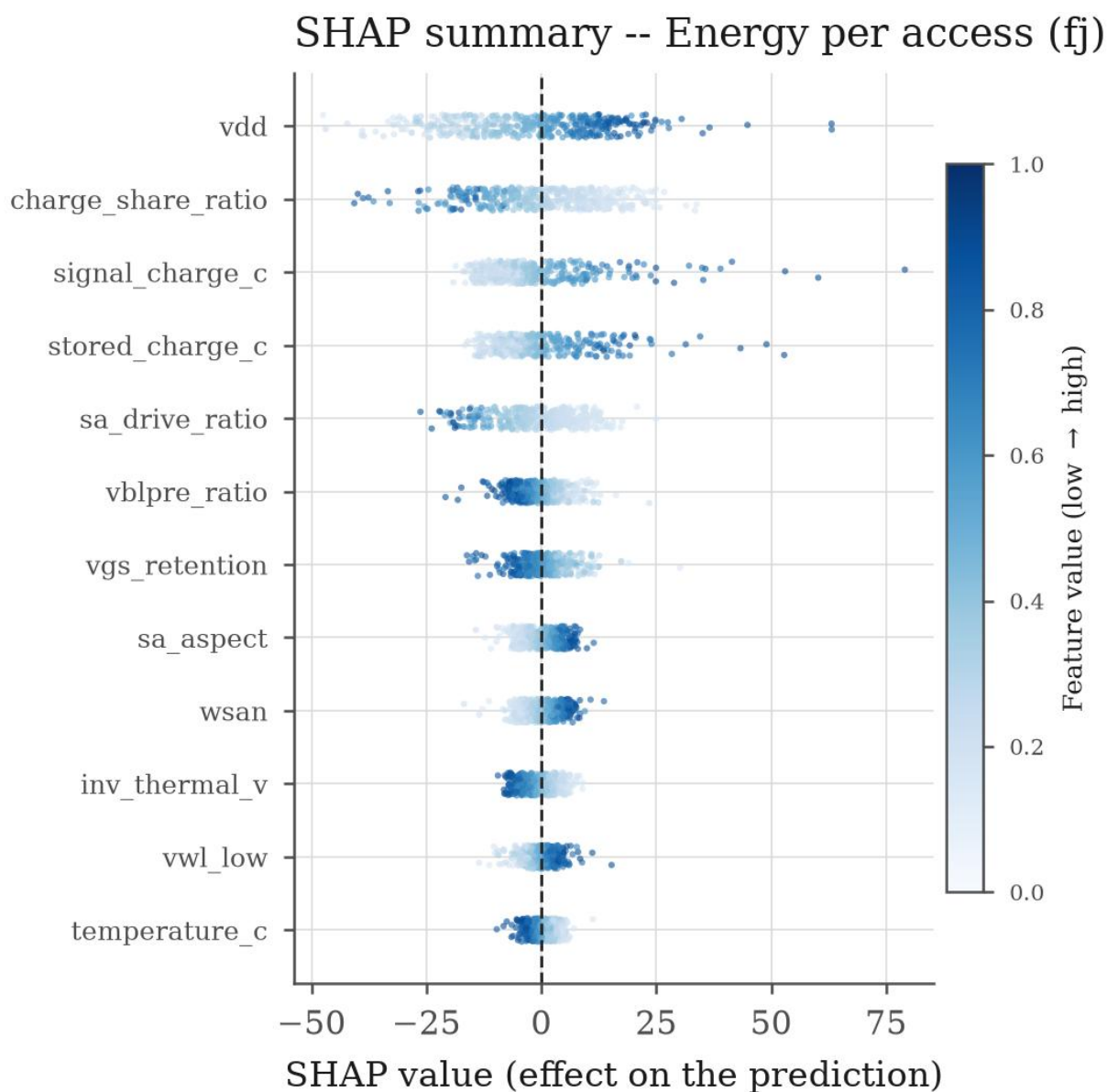


Figure 15. SHAP summary: direction and magnitude of each effect.

Feature importance -- Energy per access (fj)

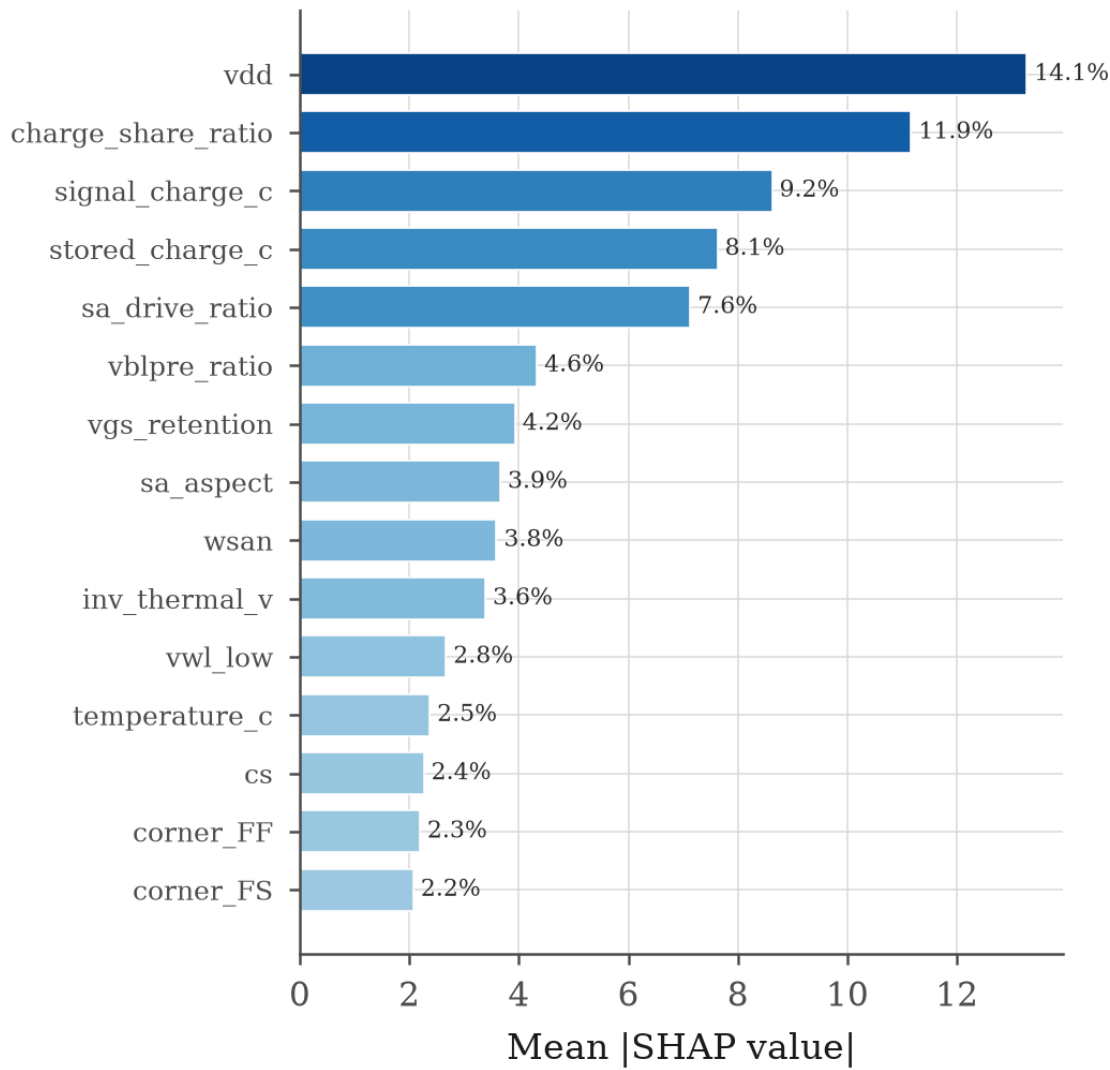


Figure 16. Global SHAP feature importance.

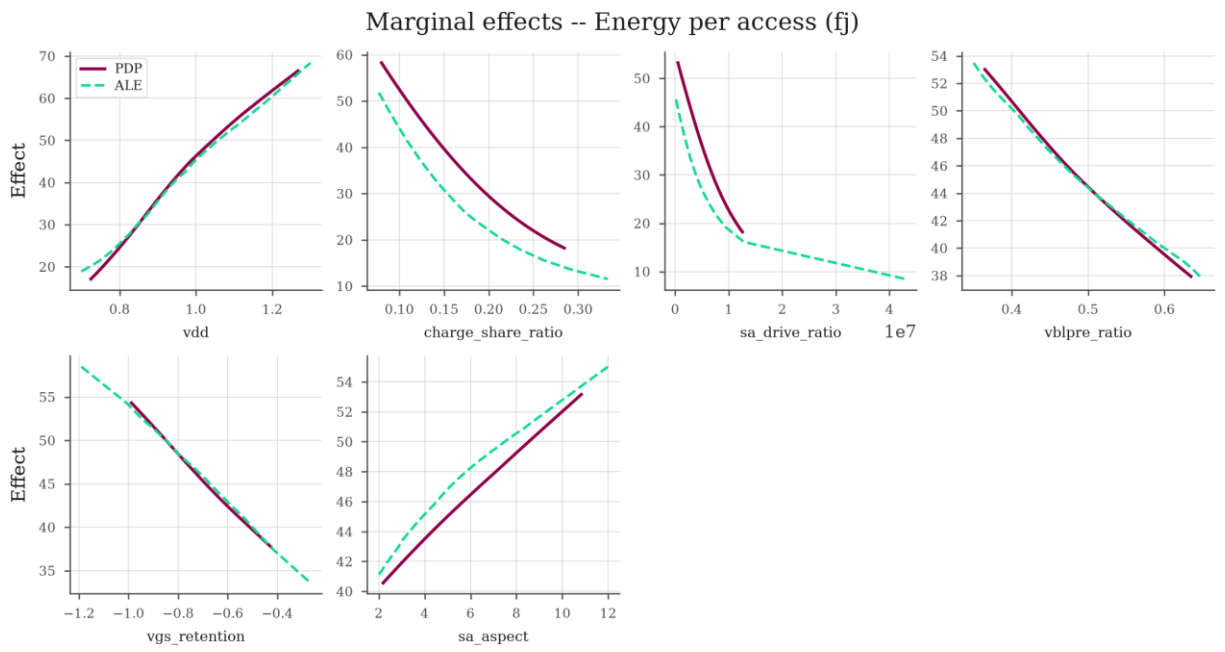


Figure 17. Marginal effects (partial dependence with ALE overlay).

Sobol' variance-based sensitivity ($S_T - S_1$ is the interaction share)

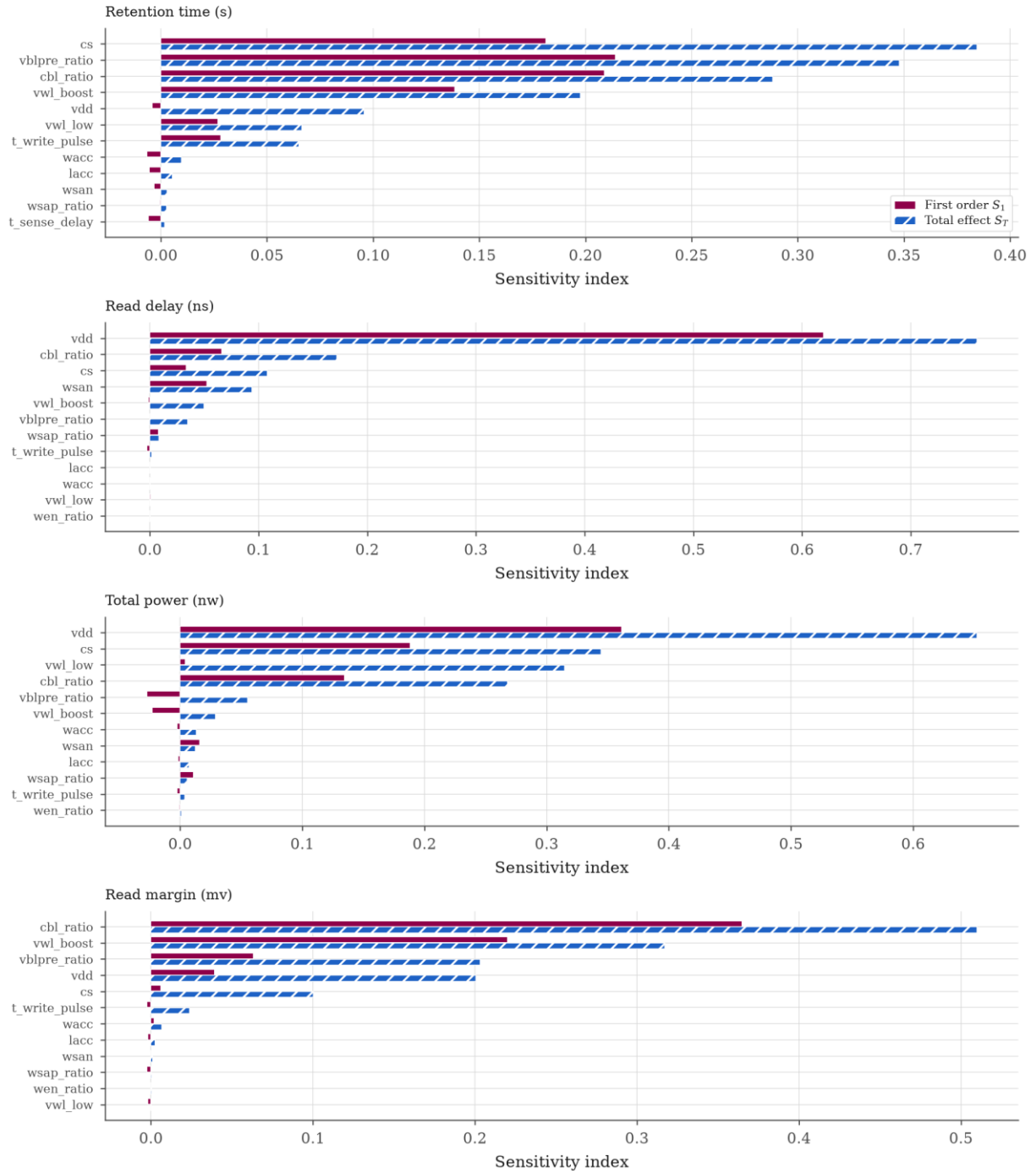


Figure 18. Sobol' first-order and total-effect sensitivity indices.

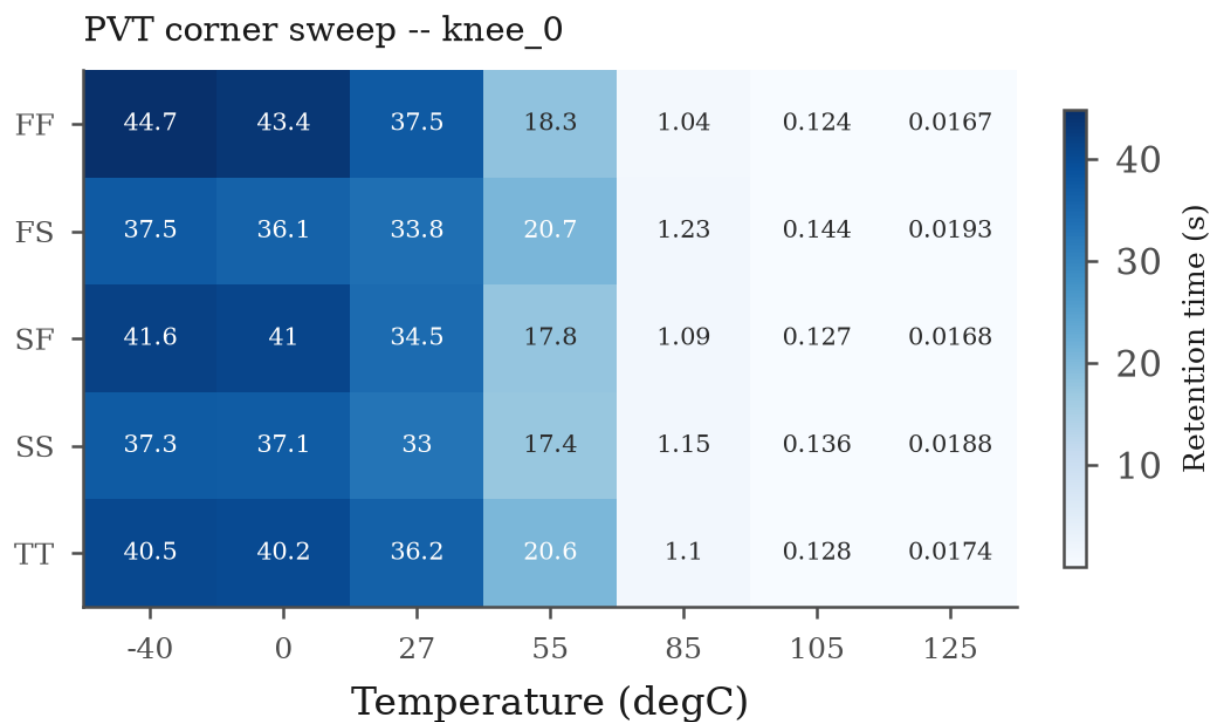


Figure 19. PVT corner sweep of a selected design.

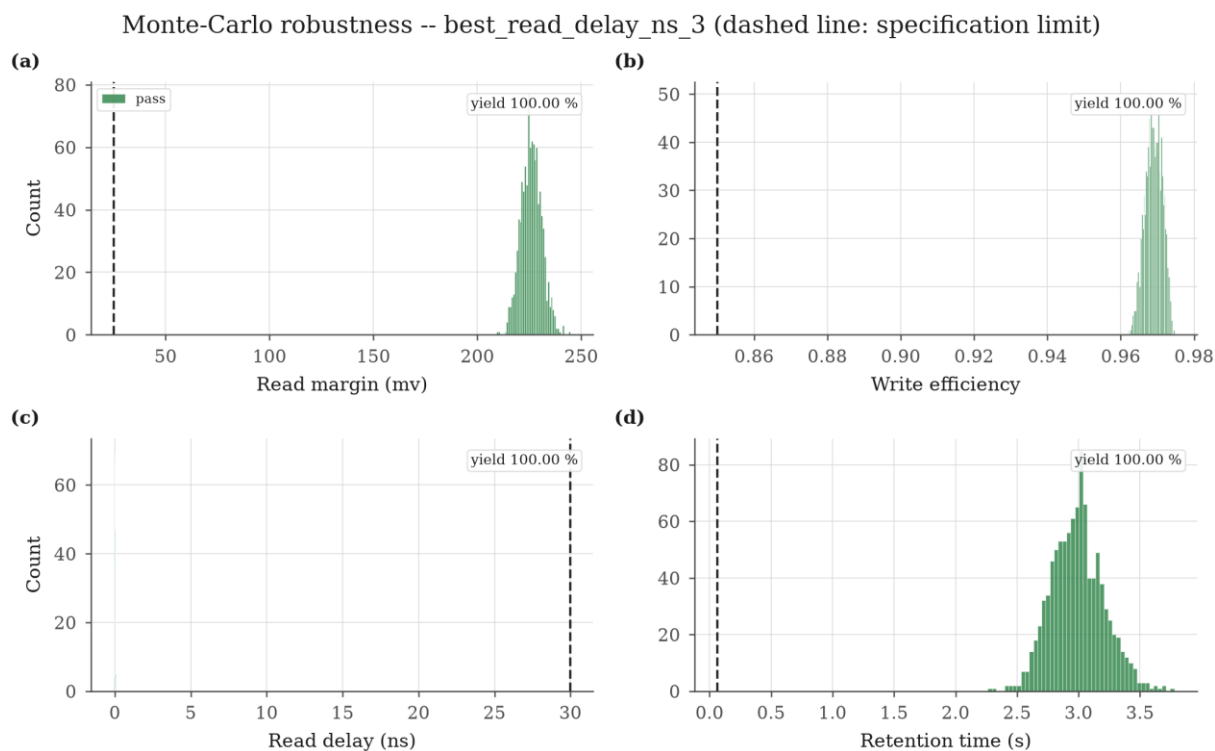


Figure 20. Monte-Carlo robustness with specification limits.

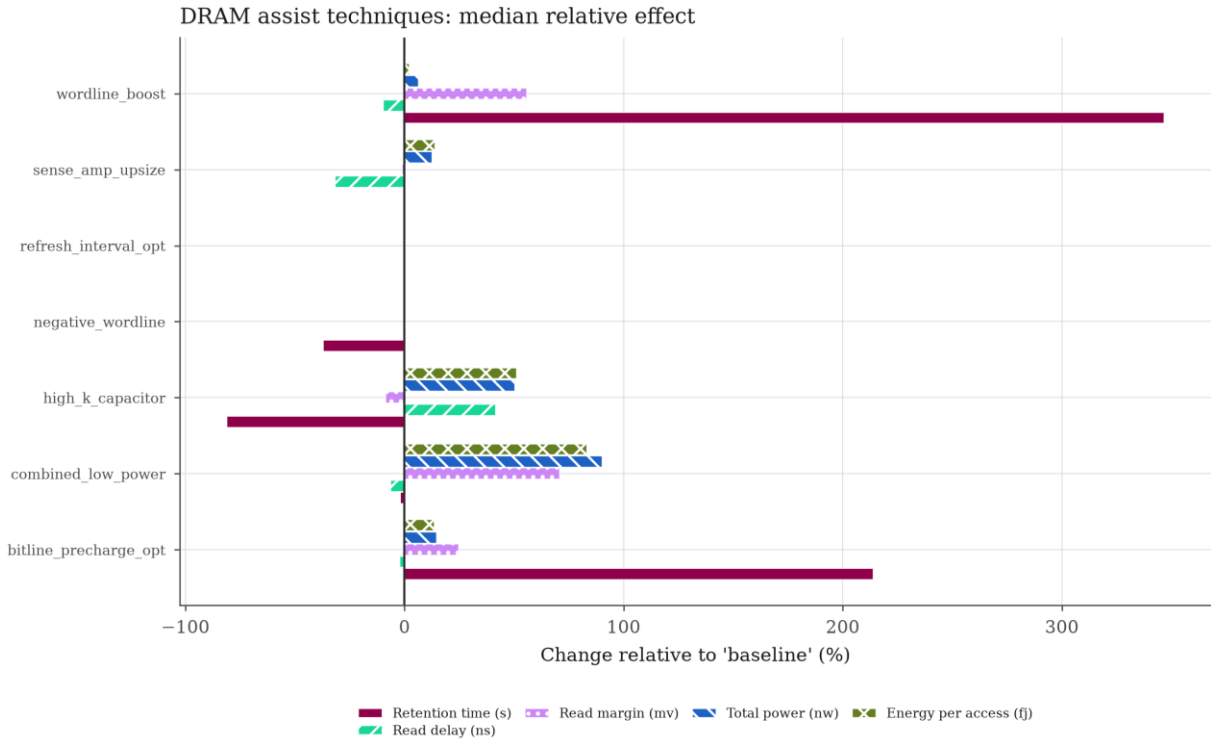


Figure 21. DRAM assist techniques: median relative effect.

Table 12. SHAP importance across responses.

feature	energyperaccess_fj	log10retentiontime_s	log10totalpower_nw	readdelay_ns	readmargin_mv	write_efficiency	meanimportance
num__vdd	14.14	8.865	18.96	21.48	11.65	2.535	12.94
num_vw/overdrive	--	9.324	3.158	--	5.538	15.95	8.494
num_signalcharge_c	9.183	--	5.060	--	--	--	7.121
num_chargeshare_ratio	11.89	--	7.108	6.464	5.121	1.899	6.496
num_vw/boost	--	8.887	--	--	5.592	4.613	6.364
num_twrite_pulse	--	6.996	--	--	3.844	7.805	6.215
num_vblpreratio	4.609	8.901	6.180	3.722	5.077	--	5.698
num_sadrive_ratio	7.592	--	5.789	3.283	--	--	5.555
num_cb/over_cs	--	6.844	--	2.230	7.402	--	5.492
num_storedcharge_c	8.120	--	5.525	2.540	--	--	5.395
num_vgsretention	4.186	2.572	5.336	7.500	6.657	--	5.250
num_cb/ratio	--	6.814	--	2.824	7.523	2.379	4.885
cat_cornerFF	2.332	2.482	4.010	5.017	4.358	10.34	4.757
cat_cornerSF	--	--	2.635	--	4.192	6.972	4.599
num__cs	2.410	4.017	2.849	8.951	--	4.262	4.498
num__wsan	3.824	--	4.620	4.952	--	4.268	4.416
cat_cornerTT	--	2.794	3.112	--	3.961	7.187	4.264

cat_cornerFS	2.207	2.261	--	--	3.902	8.529	4.224
cat_cornerSS	--	--	--	--	4.069	4.356	4.213
num_vw/low	2.840	--	3.494	5.431	4.592	--	4.089

Blank entries mean the feature did not enter that response's top-15 ranking.

Table 13. Monte-Carlo yield of the selected designs.

design_id	selection	yieldreadmargin_ mv	yieldwriteefficie ncy	yieldreaddelay _ns	yieldretentiontim e_s	yield_joi nt
knee_0	knee	100	100	100	100	100
weighted_1	weighted	100	100	100	100	100
bestretentiontim es2	bestretentiontim e_s	100	100	100	100	100
bestreaddelayns3	bestreaddelay_ns	100	100	100	100	100
besttotalpowern w4	besttotalpower_ nw	100	100	100	100	100
bestreadmargin mv5	bestreadmargin_ mv	100	100	100	100	100

yieldjoint is the fraction of trials meeting every specification simultaneously, not the product of the individual yields._

Table 14. Worst-case PVT performance.

des ign _id	rea dm argi nm v_w orst	rea dm argi nmv _ma rgin	read mar ginm vwor stcor ner	read marg inmv wors ttem p_c	wri tee ffici ency wors t cor ner	writ eeff icie ncy mar gin	write effici ency wors t_cor ner	writ eeffi cienc ywor stte mpc	rea dd ela y ns _w orst t	rea dde lay ns_ ma rgin	read dela y ns wor stco rner	read dela y ns wor stte mp_ c	rete ntio nti mes _w orst	rete ntio nti mes _ma rgin	rete ntio ntim esw orstc orne r	rete ntion time swor stte mp_ c	pa sse sal lco rn ers
kne e_0	245 .9	220. 9	SS	125	0.9 679	0.1 179	SF	125	4.9 97	25. 00	SS	125	0.0 167 1	- 0.04 729	FF	125	no
wei ght ed_ 1	230 .2	205. 2	SF	125	0.9 638	0.1 138	SF	125	2.9 93	27. 01	SS	125	0.0 173 8	- 0.04 662	SF	125	no
bes tret ent ion tim es2	260 .2	235. 2	SF	125	0.9 775	0.1 275	SF	125	5.8 49	24. 15	SS	125	0.0 205 0	- 0.04 350	FF	125	no
bes tre add ela y ns 3	224 .2	199. 2	FS	-40	0.9 658	0.1 158	TT	125	0.1 99 5	29. 80	SS	125	4.5 58e -4	- 0.06 354	SF	125	no
bes ttot	72. 44	47.4 4	SS	55	0.9 596	0.1 096	SS	27	27. 03	2.9 67	SS	0	2.1 64e	- 0.06	SF	125	no

alp ow ern w4													-4	378			
bes tre ad ma rgi nm v5	284 .2	259. 2	FS	-40	0.9 608	0.1 108	TT	125	1.2 33	28. 77	FS	125	3.1 70e -4	- 0.06 368	FF	125	no

Table 15. Sobol' sensitivity indices.

response	variable	S1	ST	interaction
retentiontimes	cs	0.1813	0.3844	0.2031
retentiontimes	vblpre_ratio	0.2141	0.3479	0.1338
retentiontimes	cbl_ratio	0.2088	0.2883	0.07943
retentiontimes	vwI_boost	0.1384	0.1977	0.05929
retentiontimes	vdd	-0.004126	0.09581	0.09994
retentiontimes	vwI_low	0.02684	0.06639	0.03955
retentiontimes	twIwritepulse	0.02829	0.06508	0.03679
retentiontimes	wacc	-0.006677	0.009693	0.01637
retentiontimes	lacc	-0.005466	0.005441	0.01091
retentiontimes	wsan	-0.003113	0.002814	0.005927
retentiontimes	wsap_ratio	-4.605e-4	0.002716	0.003176
retentiontimes	tsensedelay	-0.005824	0.001719	0.007543
retentiontimes	wen_ratio	-0.005061	0.001633	0.006694
readdelayns	vdd	0.6197	0.7608	0.1412
readdelayns	cbl_ratio	0.06615	0.1722	0.1060
readdelayns	cs	0.03327	0.1082	0.07495
readdelayns	wsan	0.05227	0.09406	0.04178
readdelayns	vwI_boost	-0.001565	0.04998	0.05154
readdelayns	vblpre_ratio	-1.808e-4	0.03477	0.03495
readdelayns	wsap_ratio	0.007755	0.008498	7.430e-4
readdelayns	twIwritepulse	-0.002909	0.001667	0.004575
readdelayns	lacc	1.063e-4	5.544e-4	4.481e-4
readdelayns	wacc	5.884e-4	4.605e-4	0
readdelayns	vwI_low	7.424e-4	4.396e-4	0
readdelayns	wen_ratio	1.238e-4	4.138e-4	2.900e-4
readdelayns	tsensedelay	-7.915e-4	2.875e-4	0.001079
totalpowernw	vdd	0.3614	0.6522	0.2908
totalpowernw	cs	0.1883	0.3444	0.1561
totalpowernw	vwI_low	0.004153	0.3148	0.3107
totalpowernw	cbl_ratio	0.1346	0.2679	0.1333
totalpowernw	vblpre_ratio	-0.02740	0.05535	0.08274
totalpowernw	vwI_boost	-0.02303	0.02883	0.05185
totalpowernw	wacc	-0.002682	0.01325	0.01593
totalpowernw	wsan	0.01580	0.01232	0
totalpowernw	lacc	-0.001696	0.007411	0.009107
totalpowernw	wsap_ratio	0.01099	0.005609	0

totalpowernw	twwritepulse	-0.002658	0.003762	0.006420
totalpowernw	wen_ratio	-0.001124	8.805e-4	0.002004
totalpowernw	tsensedelay	1.108e-4	5.662e-4	4.554e-4
readmarginmv	cbl_ratio	0.3650	0.5099	0.1449
readmarginmv	vwboost	0.2201	0.3173	0.09717
readmarginmv	vblpre_ratio	0.06325	0.2031	0.1399
readmarginmv	vdd	0.03910	0.2006	0.1615
readmarginmv	cs	0.006200	0.1002	0.09403
readmarginmv	twwritepulse	-0.002541	0.02397	0.02651
readmarginmv	wacc	0.001916	0.006645	0.004729
readmarginmv	lacc	-0.001892	0.002452	0.004344
readmarginmv	wsan	-3.742e-5	0.001032	0.001070
readmarginmv	wsap_ratio	-0.002617	5.567e-4	0.003174
readmarginmv	wen_ratio	-1.310e-4	3.589e-4	4.899e-4
readmarginmv	vwlow	-0.001813	3.054e-4	0.002119
readmarginmv	tsensedelay	-0.001700	1.052e-4	0.001805

ST - S1 is the share of variance a variable contributes through interactions rather than on its own.

Table 16. Assist-technique comparison.

assist_config	retention times	readdelayns	writedelayns	readmarginmv	totalpowernw	energyperaccess_fj	leakagecurrentfa	refreshintervalms	feasible_pct
baseline	0.2865	5.829	1.224	39.72	3.040	28.65	1.509	301.7	26.33
bitlineprechargeopt	0.8992	5.706	1.264	49.58	3.486	32.54	1.515	531.8	25.33
combinedlowpower	0.2812	5.468	1.359	67.90	5.784	52.51	486.4	151.2	74
highkcapacitor	0.05399	8.256	1.524	36.31	4.574	43.31	19.58	69.27	17
negative_wordline	0.1806	5.835	1.222	39.72	3.039	28.63	18.18	212.1	25.33
refreshintervalopt	0.2865	5.829	1.224	39.72	3.040	28.65	1.509	301.7	26.33
senseampupsize	0.2865	3.989	1.225	39.28	3.420	32.66	1.508	301.7	28.00
wordline_boost	1.280	5.266	0.8842	61.93	3.233	29.26	2.674	675.5	76.33

Paired study: every technique is applied to the identical set of 300 base designs and simulated in NGSpice.

Table 17. Paired significance tests of the assist techniques.

metric	group_1	group_2	n_pairs	statistic	p_value	median_1	median_2	cliffs_delta	effect_magnitude	a12	pvalue_holm	significant
retentiontimes	baseline	bitlineprechargeopt	300	0	2.418e-47	0.2865	0.8992	-0.2034	small	0.3983	1.451e-46	yes
retentiontimes	baseline	combinedlowpower	300	13647	4.143e-7	0.2865	0.2812	0.01107	negligible	0.5055	8.285e-7	yes
retentiontimes	baseline	highkcapacitor	300	1549	1.130e-34	0.2865	0.05399	0.3107	small	0.6553	3.390e-34	yes
retentiontimes	baseline	negative_wordline	300	61	4.118	0.286	0.180	0.0593	negligible	0.52	2.059e-	yes

mes	ine	rdline			e-42	5	6	8		97	41	
retentionti mes	basel ine	refreshinter valopt	300	--	1	0.286 5	0.286 5	0	negligible	0.50 00	1	no
retentionti mes	basel ine	senseampup size	300	171	1.565 e-40	0.286 5	0.286 5	0.0027 33	negligible	0.50 14	6.262e- 40	yes
retentionti mes	basel ine	wordline_bo ost	300	0	1.805 e-49	0.286 5	1.280	- 0.3052	small	0.34 74	1.263e- 48	yes
readdelays	basel ine	bitlineprech argeopt	192	3031	6.252 e-16	4.665	4.614	0.0441 1	negligible	0.52 21	2.501e- 15	yes
readdelays	basel ine	combined/o wpower	192	4405	2.934 e-10	4.665	4.332	0.0731 9	negligible	0.53 66	5.869e- 10	yes
readdelays	basel ine	highkcapacit or	192	0	2.943 e-33	4.665	8.256	- 0.2723	small	0.36 39	2.060e- 32	yes
readdelays	basel ine	negative_wo rdline	192	3910	3.788 e-12	4.665	4.660	0.0033 64	negligible	0.50 17	1.136e- 11	yes
readdelays	basel ine	refreshinter valopt	192	--	1	4.665	4.665	0	negligible	0.50 00	1	no
readdelays	basel ine	senseampup size	192	0	2.943 e-33	4.665	2.973	0.2536	small	0.62 68	2.060e- 32	yes
readdelays	basel ine	wordline_bo ost	192	0	2.943 e-33	4.665	4.003	0.1010	negligible	0.55 05	2.060e- 32	yes
writedelayn s	basel ine	bitlineprech argeopt	300	3	6.269 e-51	1.224	1.264	- 0.0406 0	negligible	0.47 97	4.389e- 50	yes
writedelayn s	basel ine	combined/o wpower	300	8100	6.214 e-22	1.224	1.359	- 0.1321	negligible	0.43 39	1.864e- 21	yes
writedelayn s	basel ine	highkcapacit or	300	271	9.063 e-50	1.224	1.524	- 0.2210	small	0.38 95	5.438e- 49	yes
writedelayn s	basel ine	negative_wo rdline	300	1108 9	2.202 e-14	1.224	1.222	- 0.0061 11	negligible	0.49 69	4.404e- 14	yes
writedelayn s	basel ine	refreshinter valopt	300	--	1	1.224	1.224	0	negligible	0.50 00	1	no
writedelayn s	basel ine	senseampup size	300	797	1.563 e-47	1.224	1.225	- 0.0042 89	negligible	0.49 79	7.817e- 47	yes
writedelayn s	basel ine	wordline_bo ost	300	7661	3.481 e-23	1.224	0.884 2	0.2477	small	0.62 38	1.392e- 22	yes
readmargin mv	basel ine	bitlineprech argeopt	300	0	6.084 e-51	39.72	49.58	- 0.2450	small	0.37 75	4.258e- 50	yes
readmargin mv	basel ine	combined/o wpower	300	35	8.639 e-51	39.72	67.90	- 0.5309	large	0.23 46	4.258e- 50	yes
readmargin mv	basel ine	highkcapacit or	300	15	7.071 e-51	39.72	36.31	0.1170	negligible	0.55 85	4.258e- 50	yes
readmargin mv	basel ine	negative_wo rdline	300	1278 6	7.512 e-11	39.72	39.72	- 0.0018 44	negligible	0.49 91	1.502e- 10	yes
readmargin mv	basel ine	refreshinter valopt	300	--	1	39.72	39.72	0	negligible	0.50 00	1	no
readmargin mv	basel ine	senseampup size	300	0	6.083 e-51	39.72	39.28	0.0167 2	negligible	0.50 84	4.258e- 50	yes
readmargin mv	basel ine	wordline_bo ost	300	0	6.084 e-51	39.72	61.93	- 0.4746	large	0.26 27	4.258e- 50	yes

totalpower nw	baseline	bitlineprechargeopt	300	858	2.820e-47	3.040	3.486	-0.07580	negligible	0.4621	1.410e-46	yes
totalpower nw	baseline	combined/owpower	300	196	4.305e-50	3.040	5.784	-0.3601	medium	0.3200	2.583e-49	yes
totalpower nw	baseline	highkcapacitor	300	2589	2.618e-40	3.040	4.574	-0.1991	small	0.4004	1.047e-39	yes
totalpower nw	baseline	negative_wordline	300	17290	4.405e-4	3.040	3.039	0.001733	negligible	0.5009	8.810e-4	yes
totalpower nw	baseline	refreshintervalopt	300	--	1	3.040	3.040	0	negligible	0.5000	1	no
totalpower nw	baseline	senseampupsize	300	0	6.084e-51	3.040	3.420	-0.08831	negligible	0.4558	4.258e-50	yes
totalpower nw	baseline	wordline_bios	300	14413	5.706e-8	3.040	3.233	-0.03324	negligible	0.4834	1.712e-7	yes
energyperformance_fj	baseline	bitlineprechargeopt	300	315	1.401e-49	28.65	32.54	-0.07740	negligible	0.4613	7.005e-49	yes
energyperformance_fj	baseline	combined/owpower	300	212	5.047e-50	28.65	52.51	-0.3418	medium	0.3291	3.028e-49	yes
energyperformance_fj	baseline	highkcapacitor	300	2286	1.737e-41	28.65	43.31	-0.2047	small	0.3977	6.946e-41	yes
energyperformance_fj	baseline	negative_wordline	300	7177	1.316e-24	28.65	28.63	0.003222	negligible	0.5016	3.948e-24	yes
energyperformance_fj	baseline	refreshintervalopt	300	--	1	28.65	28.65	0	negligible	0.5000	1	no
energyperformance_fj	baseline	senseampupsize	300	0	6.084e-51	28.65	32.66	-0.08038	negligible	0.4598	4.258e-50	yes
energyperformance_fj	baseline	wordline_bios	300	8254	1.674e-21	28.65	29.26	-0.003533	negligible	0.4982	3.348e-21	yes
leakagecurrentfa	baseline	bitlineprechargeopt	300	18349	0.004949	1.509	1.515	4.667e-4	negligible	0.5002	0.009899	yes
leakagecurrentfa	baseline	combined/owpower	300	7	6.526e-51	1.509	486.4	-0.7191	large	0.1404	4.258e-50	yes
leakagecurrentfa	baseline	highkcapacitor	300	8	6.592e-51	1.509	19.58	-0.4765	large	0.2617	4.258e-50	yes
leakagecurrentfa	baseline	negative_wordline	300	14618	1.214e-7	1.509	18.18	-0.1968	small	0.4016	3.641e-7	yes
leakagecurrentfa	baseline	refreshintervalopt	300	--	1	1.509	1.509	0	negligible	0.5000	1	no
leakagecurrentfa	baseline	senseampupsize	300	149	3.903e-49	1.509	1.508	0.003211	negligible	0.5016	1.561e-48	yes
leakagecurrentfa	baseline	wordline_bios	300	0	6.084e-51	1.509	2.674	-0.2034	small	0.3983	4.258e-50	yes
refreshintervalms	baseline	bitlineprechargeopt	221	0	5.197e-38	424.3	740.9	-0.1745	small	0.4128	3.638e-37	yes
refreshintervalms	baseline	combined/owpower	221	4823	5.250e-15	424.3	190.1	0.2203	small	0.6102	1.050e-14	yes
refreshintervalms	baseline	highkcapacitor	221	1329	1.441	424.3	69.27	0.4366	medium	0.71	4.323e-	yes

<i>valms</i>	ine	or			e-30					83	30	
refreshinter valms	basel ine	negative_wo rdline	221	53	1.068 e-37	424.3	274.5	0.0966 2	negligible	0.54 83	5.341e- 37	yes
refreshinter valms	basel ine	refreshinter valopt	221	--	1	424.3	424.3	0	negligible	0.50 00	1	no
refreshinter valms	basel ine	senseampup size	221	154	4.260 e-36	424.3	424.1	0.0043 61	negligible	0.50 22	1.704e- 35	yes
refreshinter valms	basel ine	wordline_bo ost	221	0	5.197 e-38	424.3	790.4	- 0.2059	small	0.39 70	3.638e- 37	yes

9. Limitations and threats to validity

These are stated plainly because they bound what the numbers mean.

1. **Device models are logic models, not DRAM models.** Every simulation uses the

public PTM BSIM4 cards for a ? nm *logic* process. Real DRAM access devices are engineered differently -- notably much stronger GIDL at the storage node -- so the *absolute* retention times here are longer than a production part would show. Relative trends, the ranking of design variables and the methodology are unaffected; absolute retention values should not be read as predictions about any commercial device.

2. **Process corners are synthesised, not signed off.** SS/FF/SF/FS are first-order

vth0/u0/tox shifts applied to the typical card, with the shift magnitudes declared in configuration. They are a reasonable spread, not foundry corners.

3. **Retention is quasi-static.** The model integrates the charge-loss ODE over a

measured leakage characteristic and is exact for an isolated capacitor whose leakage depends only on the instantaneous node voltage. Its agreement with direct transient simulation is quantified in Section 5. It does not model variable-retention-time behaviour or trap-assisted transients, which are time-dependent phenomena outside a quasi-static description.

4. **The bitline is lumped.** Cb1 is a lumped capacitance rather than a

distributed RC line with the other cells attached, so bitline resistance and neighbour coupling are not modelled.

5. **Power is reported per simulated slice.** The slice is one cell plus its column

periphery, and the active-power component depends on the configured `access_rate_hz`. All three components are reported separately so results can be reweighted for a different access profile without re-simulating.

6. **The optimiser searches the surrogate, and that shows.** A multi-objective search drives towards the edges of the design space, which is exactly where a surrogate has least support -- and any region in which it over-predicts will be exploited. The effect is measurable here. Against the held-out test split the twin reaches $R^2 = 0.95\text{--}0.999$, but on the optimiser-selected Pareto designs the median twin-versus-SPICE error is 3.0 % (*readmarginmv*), 10.3 % (*totalpowernw*), 30.8 % (*readdelayns*), 46.3 % (*retentiontimes*). Read margin and power -- the smoothest responses -- stay accurate; read delay and retention, which span decades and turn sharply near the feasibility boundary, degrade most. Expressed on the scale the quantity actually lives on, the retention discrepancy is 0.169 decades, a factor of 1.48. Two things make

this manageable rather than fatal. First it is *measured*: every selected design is returned to NGSpice, so the error is a reported quantity, not an assumption. Second, the constraints survive the round trip -- 100 % of the selected designs pass both the read and the write check under full circuit simulation. The practical reading is that the twin is a search accelerator whose proposals must be confirmed in SPICE, not a replacement for it; re-simulating the Pareto region and refitting is the natural next iteration, and is what the framework's feedback path exists for.

7. **A single technology node was run end-to-end.** Configurations for 32 nm and

22 nm ship with the framework and the pipeline is node-agnostic, but the results presented here are for the ? nm node.

10. Reproduction

```
python -m venv .venv && .venv/Scripts/activate      # Windows
pip install -e .
python scripts/00_setup_environment.py              # NGSpice + PTM models
python scripts/run_all.py --tag main                # the whole pipeline
```

Determinism: master seed 20260723, configuration hash e22061fdb47e16e7, model-card fingerprint ?. Every stage writes a manifest to logs/manifest_*.json recording its seeds, inputs, outputs, package versions and wall-clock time. See REPRODUCIBILITY.md for the full protocol and supplementary/environment_lock.txt for the exact package versions used.

References

- [1] "Re-Evaluating the Real-System Modeling Accuracy of Ramulator 2.0", arXiv, 2026 <<https://arxiv.org/abs/2606.14566>>
- [2] "Analog Design and Machine Learning: A Review", Electronics (MDPI), 2025 <<https://doi.org/10.3390/electronics14173541>>
- [3] "EasyDRAM: An FPGA-based Infrastructure for Fast and Accurate End-to-End Evaluation of Emerging DRAM Techniques", arXiv, 2025 <<https://arxiv.org/abs/2506.10441>>
- [4] "Multi-Fidelity Surrogate Models for Accelerated Multi-Objective Analog Circuit Design and Optimization", Electronics (MDPI), 2025 <<https://doi.org/10.3390/electronics15010105>>
- [5] "Analog circuit sizing based on Evolutionary Algorithms and deep learning", Expert Systems with Applications (Elsevier), 2024 <<https://www.sciencedirect.com/science/article/abs/pii/S0957417423019826>>
- [6] "Machine Learning Driven Global Optimisation Framework for Analog Circuit Design", arXiv, 2024 <<https://arxiv.org/abs/2404.02911>>
- [7] "PENDRAM: Enabling High-Performance and Energy-Efficient Processing of Deep Neural Networks through a Generalized DRAM Data Mapping Policy", arXiv, 2024 <<https://arxiv.org/abs/2408.02412>>
- [8] "PySpice-Simulated In Situ Learning with Memristor Emulation for Single-Layer Spiking Neural Networks", Electronics (MDPI) 13(23), 4665, 2024 <<https://www.mdpi.com/2079-9292/13/23/4665>>
- [9] "SPICEPilot: Navigating SPICE Code Generation and Simulation with AI Guidance", arXiv, 2024 <<https://arxiv.org/abs/2410.20553>>

- [10] "The Dawn of AI-Native EDA: Opportunities and Challenges of Large Circuit Models", arXiv, 2024 <<https://arxiv.org/abs/2403.07257>>
- [11] "When Device Modeling Meets Machine Learning: Opportunities and Challenges", ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD), 2024 <<https://dl.acm.org/doi/10.1145/3670474.3685972>>
- [12] "Performance Evaluation of Evolutionary Algorithms for Analog Integrated Circuit Design Optimisation", arXiv, 2023 <<https://arxiv.org/abs/2310.12440>>
- [13] "Python Framework for Modular and Parametric SPICE Netlists Generation", arXiv, 2023 <<https://arxiv.org/abs/2306.12224>>
- [14] "Digital Twin for Secure Semiconductor Lifecycle Management: Prospects and Applications", arXiv, 2022 <<https://arxiv.org/abs/2205.10962>>
- [15] D. W. Apley and J. Zhu, "Visualizing the Effects of Predictor Variables in Black Box Supervised Learning Models", Journal of the Royal Statistical Society Series B, vol. 82, no. 4, pp. 1059-1086, 2020 <<https://doi.org/10.1111/rssb.12377>>
- [16] J. Blank and K. Deb, "pymoo: Multi-Objective Optimization in Python", IEEE Access, vol. 8, pp. 89497-89509, 2020 <<https://doi.org/10.1109/ACCESS.2020.2990567>>
- [17] S. M. Lundberg et al., "From Local Explanations to Global Understanding with Explainable AI for Trees", Nature Machine Intelligence, vol. 2, pp. 56-67, 2020 <<https://doi.org/10.1038/s42256-019-0138-9>>
- [18] T. Akiba, S. Sano, T. Yanase, T. Ohta and M. Koyama, "Optuna: A Next-generation Hyperparameter Optimization Framework", Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2019 <<https://doi.org/10.1145/3292500.3330701>>
- [19] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush and A. Gulin, "CatBoost: unbiased boosting with categorical features", Advances in Neural Information Processing Systems (NeurIPS) 31, 2018 <<https://arxiv.org/abs/1706.09516>>
- [20] G. Ke et al., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree", Advances in Neural Information Processing Systems (NeurIPS) 30, 2017 <<https://papers.nips.cc/paper/6907-lightgbm-a-highly-efficient-gradient-boosting-decision-tree>>
- [21] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions", Advances in Neural Information Processing Systems (NeurIPS) 30, 2017 <<https://arxiv.org/abs/1705.07874>>
- [22] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System", Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016 <<https://doi.org/10.1145/2939672.2939785>>
- [23] K. Deb and H. Jain, "An Evolutionary Many-Objective Optimization Algorithm Using Reference-Point-Based Nondominated Sorting Approach, Part I", IEEE Transactions on Evolutionary Computation, vol. 18, no. 4, pp. 577-601, 2014 <<https://doi.org/10.1109/TEVC.2013.2281535>>
- [24] J. Bergstra, R. Bardenet, Y. Bengio and B. Kegl, "Algorithms for Hyper-Parameter Optimization", Advances in Neural Information Processing Systems (NeurIPS) 24, 2011 <<https://papers.nips.cc/paper/4443-algorithms-for-hyper-parameter-optimization>>

- [25] A. Saltelli et al., "Variance based sensitivity analysis of model output: Design and estimator for the total sensitivity index", *Computer Physics Communications*, vol. 181, no. 2, pp. 259-270, 2010
<<https://doi.org/10.1016/j.cpc.2009.09.018>>
- [26] M. Okabe and K. Ito, "Color Universal Design - How to make figures and presentations that are friendly to colorblind people", 2008 <<https://jfly.uni-koeln.de/color/>>
- [27] Q. Zhang and H. Li, "MOEA/D - A Multiobjective Evolutionary Algorithm Based on Decomposition", *IEEE Transactions on Evolutionary Computation*, vol. 11, no. 6, pp. 712-731, 2007
<<https://doi.org/10.1109/TEVC.2007.892759>>
- [28] J. Demsar, "Statistical Comparisons of Classifiers over Multiple Data Sets", *Journal of Machine Learning Research*, vol. 7, pp. 1-30, 2006 <<https://www.jmlr.org/papers/v7/demsar06a.html>>
- [29] W. Zhao and Y. Cao, "New Generation of Predictive Technology Model for Sub-45 nm Early Design Exploration", *IEEE Transactions on Electron Devices*, vol. 53, no. 11, pp. 2816-2823, 2006
<<https://doi.org/10.1109/TED.2006.884077>>
- [30] K. Deb, A. Pratap, S. Agarwal and T. Meyarivan, "A Fast and Elitist Multiobjective Genetic Algorithm - NSGA-II", *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182-197, 2002
<<https://doi.org/10.1109/4235.996017>>
- [31] R. Storn and K. Price, "Differential Evolution - A Simple and Efficient Heuristic for Global Optimization", *Journal of Global Optimization*, vol. 11, pp. 341-359, 1997 <<https://doi.org/10.1023/A:1008202821328>>
- [32] J. Kennedy and R. Eberhart, "Particle Swarm Optimization", *Proceedings of ICNN'95 - International Conference on Neural Networks*, pp. 1942-1948, 1995 <<https://doi.org/10.1109/ICNN.1995.488968>>
- [33] M. J. M. Pelgrom, A. C. J. Duinmaijer and A. P. G. Welbers, "Matching Properties of MOS Transistors", *IEEE Journal of Solid-State Circuits*, vol. 24, no. 5, pp. 1433-1439, 1989
<<https://doi.org/10.1109/JSSC.1989.572629>>
- [34] M. D. McKay, R. J. Beckman and W. J. Conover, "A Comparison of Three Methods for Selecting Values of Input Variables in the Analysis of Output from a Computer Code", *Technometrics*, vol. 21, no. 2, pp. 239-245, 1979 <<https://doi.org/10.2307/1268522>>
- [35] BSIM Group, University of California, Berkeley, "BSIM4 MOSFET Model User's Manual", UC Berkeley Device Group <<https://bsim.berkeley.edu/models/bsim4/>>
- [36] F. Salvaire, "PySpice: Simulate electronic circuits using Python and the Ngspice / Xyce simulators"
<<https://github.com/PySpice-org/PySpice>>
- [37] Ngspice development team, "Ngspice: open-source mixed-level/mixed-signal circuit simulator"
<<https://ngspice.sourceforge.io/>>